



? A steam locomotive uses the first law of thermodynamics: Water is heated and boils, and the expanding steam does work to propel the locomotive. Would it be possible for the steam to propel the locomotive by doing work as it *condenses*? (i) Yes; (ii) no; (iii) answer depends on the details of how the steam condenses.

19 The First Law of Thermodynamics

Every time you drive a gasoline-powered car, turn on an air conditioner, or cook a meal, you reap the benefits of *thermodynamics*, the study of relationships involving heat, mechanical work, and other aspects of energy and energy transfer. For example, in a car engine heat is generated by the chemical reaction of oxygen and vaporized gasoline in the engine's cylinders. The heated gas pushes on the pistons within the cylinders, doing mechanical work that is used to propel the car. This is an example of a *thermodynamic process*.

The first law of thermodynamics, central to the understanding of such processes, is an extension of the principle of conservation of energy. It broadens this principle to include energy exchange by both heat transfer and mechanical work and introduces the concept of the *internal energy* of a system. Conservation of energy plays a vital role in every area of physical science, and the first law has extremely broad usefulness. To state energy relationships precisely, we need the concept of a *thermodynamic system*. We'll discuss *heat* and *work* as two means of transferring energy into or out of such a system.

19.1 THERMODYNAMIC SYSTEMS

We have studied energy transfer through mechanical work (Chapter 6) and through heat transfer (Chapters 17 and 18). Now we are ready to combine and generalize these principles.

We always talk about energy transfer to or from some specific *system*. The system might be a mechanical device, a biological organism, or a specified quantity of material, such as the refrigerant in an air conditioner or steam expanding in a turbine. In general, a **thermodynamic system** is any collection of objects that is convenient to regard as a unit, and that may have the potential to exchange energy with its surroundings. A familiar example is a quantity of popcorn kernels in a pot with a lid. When the pot is placed on a stove, energy is added to the popcorn by conduction of heat. As the popcorn pops and expands, it does work as it exerts an upward force on the lid and moves it through a displacement (**Fig. 19.1**). The *state* of the popcorn—its volume, temperature,

LEARNING OUTCOMES

In this chapter, you'll learn...

- 19.1 The significance of thermodynamic systems and processes.
- 19.2 How to calculate work done by a system when its volume changes.
- 19.3 What is meant by a path between thermodynamic states.
- 19.4 How to interpret and use the first law of thermodynamics.
- 19.5 Four important kinds of thermodynamic processes.
- 19.6 Why the internal energy of an ideal gas depends on temperature only.
- 19.7 The difference between molar heat capacities at constant volume and at constant pressure.
- 19.8 How to analyze adiabatic processes in an ideal gas.

You'll need to review...

- 6.3 Work done by a force.
- 7.3 Internal energy.
- 17.5, 18.4 Specific heat and molar heat capacity.
- 18.1 pV -diagrams.

Figure 19.1 The popcorn in the pot is a thermodynamic system. In the thermodynamic process shown here, heat is added to the system, and the system does work on its surroundings to lift the lid of the pot.



CAUTION Be careful with the sign of work W Note that our sign rule for work is *opposite* to the one we used in mechanics, in which we always spoke of the work done by the forces acting *on* an object. In thermodynamics it is usually more convenient to call W the work done *by* the system so that when a system expands, the pressure, volume change, and work are all positive. Use the sign rules for work and heat consistently! **I**

and pressure—changes as it pops. A process such as this one, in which there are changes in the state of a thermodynamic system, is called a **thermodynamic process**.

In mechanics we used the concept of *system* with free-body diagrams and with conservation of energy and momentum. For *thermodynamic* systems, as for all others, it is essential to define clearly at the start exactly what is and is not included in the system. Only then can we describe unambiguously the energy transfers into and out of that system. For instance, in our popcorn example we defined the system to include the popcorn but not the pot, lid, or stove.

Thermodynamics has its roots in many practical problems other than popping popcorn (Fig. 19.2). The gasoline engine in an automobile, the jet engines in an airplane, and the rocket engines in a launch vehicle use the heat of combustion of their fuel to perform mechanical work in propelling the vehicle. Muscle tissue in living organisms metabolizes chemical energy in food and performs mechanical work on the organism's surroundings. A steam engine or steam turbine uses the heat of combustion of coal or other fuel to perform mechanical work such as driving an electric generator or pulling a train.

Signs for Heat and Work in Thermodynamics

We describe the energy relationships in any thermodynamic process in terms of the quantity of heat Q added *to* the system and the work W done *by* the system. Both Q and W may be positive, negative, or zero (Fig. 19.3). A positive value of Q represents heat flow *into* the system, with a corresponding input of energy to it; negative Q represents heat flow *out of* the system. A positive value of W represents work done *by* the system against its surroundings, such as work done by an expanding gas, and hence corresponds to energy *leaving* the system. Negative W , such as work done during compression of a gas in which work is done *on the gas* by its surroundings, represents energy *entering* the system. We'll use these conventions consistently in the examples in this chapter and the next.

TEST YOUR UNDERSTANDING OF SECTION 19.1 In Example 17.7 (Section 17.6), what is the sign of Q for the coffee? For the aluminum cup? If a block slides along a horizontal surface with friction, what is the sign of W for the block?

ANSWER

I negative, positive, positive Heat flows out of the coffee, so $Q_{\text{coffee}} < 0$; heat flows into the aluminum cup, so $Q_{\text{aluminum}} > 0$. In mechanics, we would say that negative work is done *on* the block, since the surface exerts a force on the block that opposes the block's motion. But in thermodynamics we use the opposite convention and say that $W > 0$, which means that positive work is done *by* the block on the surface.

Figure 19.2 (a) A rocket engine uses the heat of combustion of its fuel to do work propelling the launch vehicle. (b) Humans and other biological organisms are more complicated systems than we can analyze fully in this book, but the same basic principles of thermodynamics apply to them.

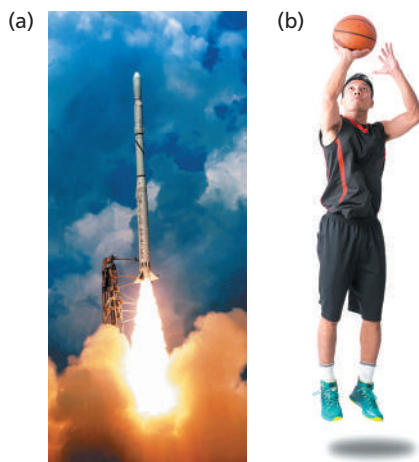
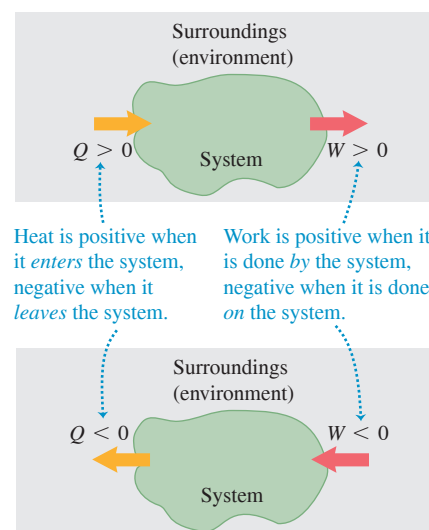


Figure 19.3 A thermodynamic system may exchange energy with its surroundings (environment) by means of heat, work, or both. Note the sign conventions for Q and W .



19.2 WORK DONE DURING VOLUME CHANGES

A simple example of a thermodynamic system is a quantity of gas enclosed in a cylinder with a movable piston. Internal-combustion engines, steam engines, and compressors in refrigerators and air conditioners all use some version of such a system. In the next several sections we'll use the gas-in-cylinder system to explore several kinds of thermodynamic processes.

We'll use a microscopic viewpoint, based on the kinetic and potential energies of individual molecules in a material, to develop intuition about thermodynamic quantities. But it is important to understand that the central principles of thermodynamics can be treated in a completely *macroscopic* way, without reference to microscopic models. Indeed, part of the great power and generality of thermodynamics is that it does *not* depend on details of the structure of matter.

First we consider the *work* done by the system during a volume change. When a gas expands, it pushes outward on its boundary surfaces as they move outward. Hence an expanding gas always does positive work. The same thing is true of any material that expands under pressure, such as the popcorn in Fig. 19.1.

We can understand the work done by a gas in a volume change by considering the molecules that make up the gas. When one such molecule collides with a stationary surface, it exerts a momentary force on the wall but does no work because the wall does not move. But if the surface is moving, like a piston in a gasoline engine, the molecule *does* do work on the surface during the collision. If the piston in **Fig. 19.4a** moves to the right, so the volume of the gas increases, the molecules that strike the piston exert a force through a distance and do *positive* work on the piston. If the piston moves toward the left as in Fig. 19.4b, so the volume of the gas decreases, positive work is done *on* the molecule during the collision. Hence the gas molecules do *negative* work on the piston.

Figure 19.5 shows a system whose volume can change (a gas, liquid, or solid) in a cylinder with a movable piston. Suppose that the cylinder has cross-sectional area A and that the pressure exerted by the system at the piston face is p . The total force F exerted by the system on the piston is $F = pA$. When the piston moves out an infinitesimal distance dx , the work dW done by this force is

$$dW = F dx = pA dx$$

But

$$A dx = dV$$

where dV is the infinitesimal change of volume of the system. Thus we can express the work done by the system in this infinitesimal volume change as

$$dW = p dV \quad (19.1)$$

In a finite change of volume from V_1 to V_2 ,

$$\text{Work done in a volume change} \rightarrow W = \int_{V_1}^{V_2} p dV \quad (19.2)$$

Upper limit = final volume
 Integral of the pressure with respect to volume
 Lower limit = initial volume

In general, the pressure of the system may vary during the volume change. For example, this is the case in the cylinders of an automobile engine as the pistons move back and forth. To evaluate the integral in Eq. (19.2), we have to know how the pressure varies as a function of volume. We can represent this relationship as a graph of p as a function of V (a pV -diagram, described at the end of Section 18.1). **Figure 19.6** (next page) shows a simple example. In this figure, Eq. (19.2) is represented graphically as the *area* under the curve of p versus V between the limits V_1 and V_2 . (In Section 6.3 we used a similar interpretation of the work done by a force F as the area under the curve of F versus x between the limits x_1 and x_2 .)

According to the rule we stated in Section 19.1, work is *positive* when a system *expands*. In an expansion from state 1 to state 2 in Fig. 19.6a, the area under the curve and

Figure 19.4 A molecule striking a piston (a) does positive work if the piston is moving away from the molecule and (b) does negative work if the piston is moving toward the molecule. Hence a gas does positive work when it expands as in (a) but does negative work when it compresses as in (b).

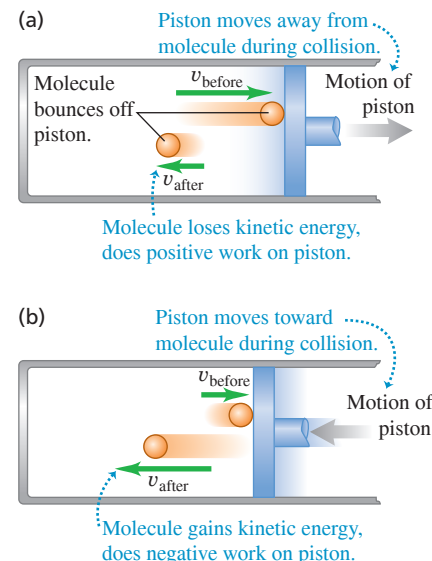


Figure 19.5 The infinitesimal work done by the system during the small expansion dx is $dW = pA dx$.

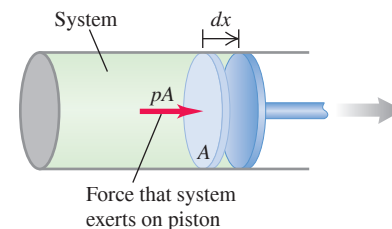
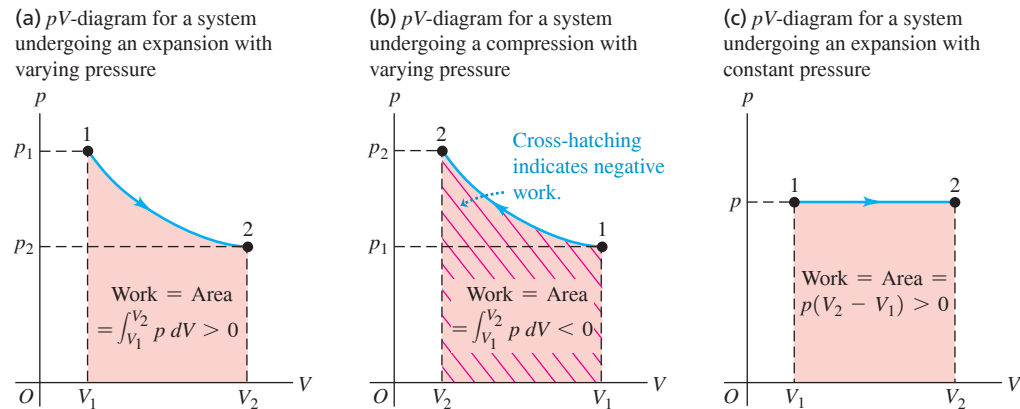


Figure 19.6 The work done equals the area under the curve on a pV -diagram.

the work are positive. A *compression* from 1 to 2 in Fig. 19.6b gives a *negative* area; when a system is compressed, its volume decreases and it does *negative* work on its surroundings (see also Fig. 19.4b).

If the pressure p remains constant while the volume changes from V_1 to V_2 (Fig. 19.6c), the work done by the system is

$$\text{Work done in a volume change at constant pressure} \quad W = p(V_2 - V_1) \quad (19.3)$$

Pressure
Final volume
Initial volume

If the volume is *constant*, there is no displacement and the system does no work.

CAUTION Be careful with subscripts 1 and 2 In Eq. (19.2), V_1 is the *initial* volume and V_2 is the *final* volume. That's why labels 1 and 2 are reversed in Fig. 19.6b compared to Fig. 19.6a, even though both processes move between the same two thermodynamic states. ■

EXAMPLE 19.1 Isothermal expansion of an ideal gas

As an ideal gas undergoes an *isothermal* (constant-temperature) expansion at temperature T , its volume changes from V_1 to V_2 . How much work does the gas do?

IDENTIFY and SET UP The ideal-gas equation, Eq. (18.3), tells us that if the temperature T of n moles of an ideal gas is constant, the quantity $pV = nRT$ is also constant. If V changes, p changes as well, so we *cannot* use Eq. (19.3) to calculate the work done. Instead we must evaluate the integral in Eq. (19.2), so we must know p as a function of V ; for this we use Eq. (18.3).

EXECUTE From Eq. (18.3),

$$p = \frac{nRT}{V}$$

We substitute this into the integral of Eq. (19.2), take the constant factor nRT outside, and evaluate the integral:

$$\begin{aligned} W &= \int_{V_1}^{V_2} p \, dV = nRT \int_{V_1}^{V_2} \frac{dV}{V} \\ &= nRT \ln \frac{V_2}{V_1} \quad (\text{ideal gas, isothermal process}) \end{aligned}$$

We can rewrite this expression for W in terms of p_1 and p_2 . Because $pV = nRT$ is constant,

$$p_1 V_1 = p_2 V_2 \quad \text{or} \quad \frac{V_2}{V_1} = \frac{p_1}{p_2}$$

so

$$W = nRT \ln \frac{p_1}{p_2} \quad (\text{ideal gas, isothermal process})$$

EVALUATE We check our result by noting that in an expansion $V_2 > V_1$ and the ratio V_2/V_1 is greater than 1. The logarithm of a number greater than 1 is positive, so $W > 0$, as it should be. As an additional check, look at our second expression for W : In an isothermal expansion the volume increases and the pressure drops, so $p_2 < p_1$, the ratio $p_1/p_2 > 1$, and $W = nRT \ln(p_1/p_2)$ is again positive.

Our result for W also applies to an isothermal *compression* of a gas, for which $V_2 < V_1$ and $p_2 > p_1$.

KEYCONCEPT To find the amount of work a system does on its surroundings during a volume change, you can either calculate the integral of pressure p with respect to volume V , or find the area under the curve of p versus V . The work is positive when the volume increases, and negative when the volume decreases.

TEST YOUR UNDERSTANDING OF SECTION 19.2 A quantity of ideal gas undergoes an expansion that doubles its volume. Does the gas do more work on its surroundings if the expansion is at constant *pressure* or at constant *temperature*? (i) Constant pressure; (ii) constant temperature; (iii) the same amount in both cases; (iv) not enough information is given to decide.

ANSWER (ii) The work done in an expansion is represented by the area under the curve of pressure p versus volume V . In an isothermal expansion the pressure decreases as the volume increases, so the pV -diagram looks like Fig. 19.6a and the work done equals the shaded area under the blue curve from point 1 to point 2. If, however, the expansion is at constant pressure, the curve of p versus V would be the same as the dashed horizontal line at pressure p_2 in Fig. 19.6a. The area under this dashed line is smaller than the area under the blue curve for an isothermal expansion, so less work is done in the constant-pressure expansion than in the isothermal expansion.

19.3 PATHS BETWEEN THERMODYNAMIC STATES

We've seen that if a thermodynamic process involves a change in volume, the system undergoing the process does work (either positive or negative) on its surroundings. Heat also flows into or out of the system during the process if there is a temperature difference between the system and its surroundings. Let's now examine how the work done by and the heat added to the system during a thermodynamic process depend on the details of how the process takes place.

Work Done in a Thermodynamic Process

When a thermodynamic system changes from an initial state to a final state, it passes through a series of intermediate states. We call this series of states a **path**. There are always infinitely many possibilities for these intermediate states. When all are equilibrium states, the path can be plotted on a pV -diagram (Fig. 19.7a). Point 1 represents an initial state with pressure p_1 and volume V_1 , and point 2 represents a final state with pressure p_2 and volume V_2 . To pass from state 1 to state 2, we could keep the pressure constant at p_1 while the system expands to volume V_2 (point 3 in Fig. 19.7b), then reduce the pressure to p_2 (probably by decreasing the temperature) while keeping the volume constant at V_2 (to point 2). The work done by the system during this process is the area under the line $1 \rightarrow 3$; no work is done during the constant-volume process $3 \rightarrow 2$. Or the system might traverse the path $1 \rightarrow 4 \rightarrow 2$ (Fig. 19.7c); then the work is the area under the line $4 \rightarrow 2$, since no work is done during the constant-volume process $1 \rightarrow 4$. The smooth curve from 1 to 2 is another possibility (Fig. 19.7d), and the work for this path is different from that for either of the other paths.

We conclude that *the work done by the system depends not only on the initial and final states, but also on the intermediate states—that is, on the path*. Furthermore, we can take the system through a series of states forming a closed loop, such as $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$. In this case the final state is the same as the initial state, but the total work done by the system is *not* zero. (In fact, it is represented on the graph by the area enclosed by the loop; see Exercise 19.7.) So we can't talk about the amount of work *contained* in a system. In a particular state, a system may have definite values of the state coordinates p , V , and T , but it wouldn't make sense to say that it has a definite value of W .

Heat Added in a Thermodynamic Process

Like work, the *heat* added to a thermodynamic system when it undergoes a change of state depends on the path from the initial state to the final state. Here's an example. Suppose we want to change the volume of a certain quantity of an ideal gas from 2.0 L to 5.0 L while keeping the temperature constant at $T = 300$ K. **Figure 19.8** (next page) shows two different ways to do this. In Fig. 19.8a the gas is contained in a cylinder with a piston, with an initial volume of 2.0 L. We let the gas expand slowly, supplying heat from the electric heater to keep the temperature at 300 K until the gas reaches its final volume of 5.0 L. The gas absorbs a definite amount of heat in this isothermal process.

Figure 19.8b shows a different process leading to the same final state. The container is surrounded by insulating walls and is divided by a thin, breakable partition into two compartments. The lower part has volume 2.0 L and the upper part has volume 3.0 L. In the lower compartment we place the same amount of the same gas as in Fig. 19.8a,

Figure 19.7 The work done by a system during a transition between two states depends on the path chosen.

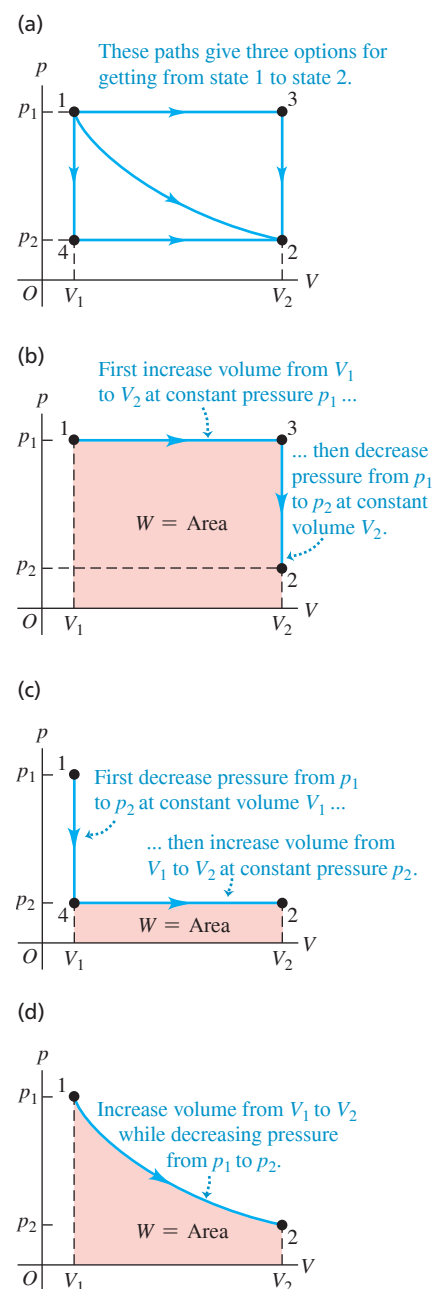
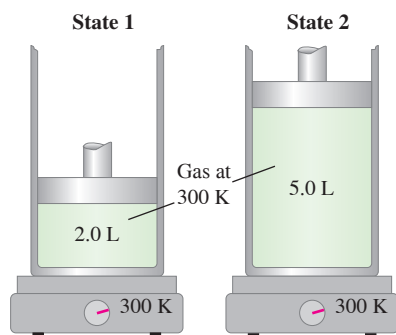
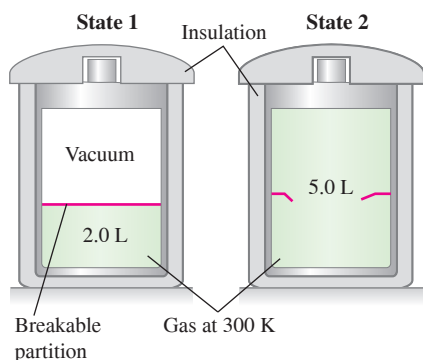


Figure 19.8 (a) Slow, controlled isothermal expansion of a gas from an initial state 1 to a final state 2 with the same temperature but lower pressure. (b) Rapid, uncontrolled expansion of the same gas starting at the same state 1 and ending at the same state 2.

(a) System does work on piston; hot plate adds heat to system ($W > 0$ and $Q > 0$).



(b) System does no work; no heat enters or leaves system ($W = 0$ and $Q = 0$).



again at $T = 300$ K. The initial state is the same as before. Now we break the partition; the gas expands rapidly, with no heat passing through the insulating walls. The final volume is 5.0 L, the same as in Fig. 19.8a. The expanding gas does no work because it doesn't push against anything that moves. This uncontrolled expansion of a gas into vacuum is called a **free expansion**; we'll discuss it further in Section 19.6.

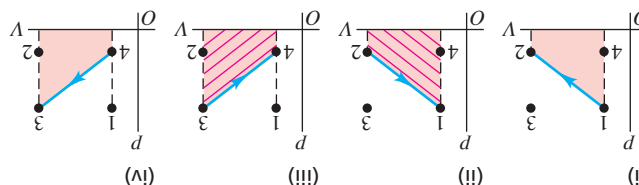
Experiments show that when an ideal gas undergoes a free expansion, there is no temperature change. So the final state of the gas is the same as in Fig. 19.8a. The intermediate pressures and volumes during the transition from state 1 to state 2 are entirely different in the two cases; Figs. 19.8a and 19.8b represent *two different paths* connecting the *same states* 1 and 2. For the path in Fig. 19.8b, *no* heat is transferred into the system, and the system does no work. Like work, *heat depends not only on the initial and final states but also on the path*.

Because of this path dependence, it would not make sense to say that a system “contains” a certain quantity of heat. To see this, suppose we assign an arbitrary value to the “heat in an object” in some reference state. Then presumably the “heat in the object” in some other state would equal the heat in the reference state plus the heat added when the object goes to the second state. But that's ambiguous, as we have just seen; the heat added depends on the *path* we take from the reference state to the second state. We are forced to conclude that there is *no* consistent way to define “heat in an object”; it is not a useful concept.

While it doesn't make sense to talk about “work in an object” or “heat in an object,” it *does* make sense to speak of the amount of *internal energy* in an object. This important concept is our next topic.

TEST YOUR UNDERSTANDING OF SECTION 19.3 The system described in Fig. 19.7a undergoes four different thermodynamic processes. Each process is represented in a pV -diagram as a straight line from the initial state to the final state. (These processes are different from those shown in the pV -diagrams of Fig. 19.7.) Rank the processes in order of the amount of work done by the system, from the most positive to the most negative. (i) $1 \rightarrow 2$; (ii) $2 \rightarrow 1$; (iii) $3 \rightarrow 4$; (iv) $4 \rightarrow 3$.

ANSWER



The accompanying figure shows the pV -diagrams for each of the four processes. The trapezoidal area under the curve, and hence the absolute value of the work, is the same in all four cases. In cases (i) and (iv) the volume increases, so the system does positive work as it expands against its surroundings. In cases (ii) and (iii) the volume decreases, so the system does negative work (shown by cross-hatching) as the surroundings push inward on it.

19.4 INTERNAL ENERGY AND THE FIRST LAW OF THERMODYNAMICS

Internal energy is one of the most important concepts in thermodynamics. In Section 7.3, when we discussed energy changes for an object sliding with friction, we stated that warming an object increased its internal energy and that cooling the object decreased its internal energy. But what *is* internal energy? We can look at it in various ways; let's start with one based on the ideas of mechanics. Matter consists of atoms and molecules, and these are made up of particles having kinetic and potential energies. We *tentatively* define the **internal energy** of a system as the sum of the kinetic energies of all of its constituent particles, plus the sum of all the potential energies of interaction among these particles.

CAUTION Is it internal? Internal energy does *not* include potential energy arising from the interaction between the system and its surroundings. If the system is a glass of water, placing it on a high shelf increases the gravitational potential energy arising from the interaction between the glass and the earth. But this has no effect on the interactions among the water molecules, and so the internal energy of the water does not change. **I**

We use the symbol U for internal energy. (We used this symbol in our study of mechanics to represent potential energy. However, U has a different meaning in thermodynamics.) During a change of state of the system, the internal energy may change from an initial value U_1 to a final value U_2 . We denote the change in internal energy as $\Delta U = U_2 - U_1$.

When we add a quantity of heat Q to a system and the system does no work during the process (so $W = 0$), the internal energy increases by an amount equal to Q ; that is, $\Delta U = Q$. When a system does work W by expanding against its surroundings and no heat is added during the process, energy leaves the system and the internal energy decreases: W is positive, Q is zero, and $\Delta U = -W$. When *both* heat transfer and work occur, the *total* change in internal energy is

$$\text{First law of thermodynamics: } \Delta U = Q - W \quad (19.4)$$

Internal energy change of thermodynamic system
Heat added to system $\cdots$ Work done by system

We can rearrange this to the form

$$Q = \Delta U + W \quad (19.5)$$

The message of Eq. (19.5) is that when heat Q is added to a system, some of this added energy remains within the system, changing its internal energy by ΔU ; the remainder leaves the system as the system does work W on its surroundings. Because W and Q may be positive, negative, or zero, ΔU can be positive, negative, or zero for different processes (Fig. 19.9).

Equation (19.4) or (19.5) is the **first law of thermodynamics**. It is a generalization of the principle of conservation of energy to include energy transfer through heat as well as mechanical work. As you'll see in later chapters, this principle can be extended to ever-broader classes of phenomena by identifying additional forms of energy and energy transfer. In every situation in which it seems that the total energy in all known forms is not conserved, it has been possible to identify a new form of energy such that the total energy, including the new form, *is* conserved.

Understanding the First Law of Thermodynamics

At the beginning of this discussion we tentatively defined internal energy in terms of microscopic kinetic and potential energies. But actually *calculating* internal energy in this way for any real system would be hopelessly complicated. Furthermore, this definition isn't an *operational* one: It doesn't describe how to determine internal energy from physical quantities that we can measure.

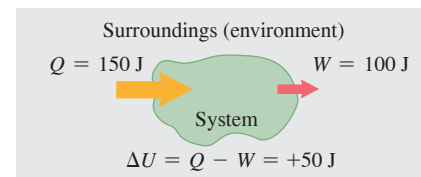
So let's look at internal energy in another way. Starting over, we define the *change* in internal energy ΔU during any change of a system as the quantity given by Eq. (19.4), $\Delta U = Q - W$. This *is* an operational definition because we can measure Q and W . It does not define U itself, only ΔU . This is not a shortcoming because we can *define* the internal energy of a system to have a specified value in some reference state, and then use Eq. (19.4) to define the internal energy in any other state. This is analogous to our treatment of potential energy in Chapter 7, in which we arbitrarily defined the potential energy of a mechanical system to be zero at a certain position.

This new definition trades one difficulty for another. If we define ΔU by Eq. (19.4), then when the system goes from state 1 to state 2 by two different paths, how do we know that ΔU is the same for the two paths? We have already seen that Q and W are, in general, *not* the same for different paths. If ΔU , which equals $Q - W$, is also path dependent, then ΔU is ambiguous. If so, the concept of internal energy of a system is subject to the same criticism as the erroneous concept of quantity of heat in a system, as we discussed at the end of Section 19.3.

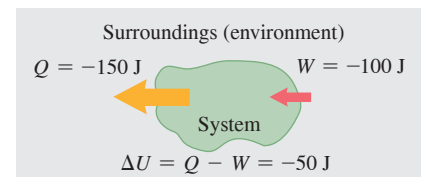
The only way to answer this question is through *experiment*. For various materials we measure Q and W for various changes of state and various paths to learn whether ΔU is or is not path dependent. The results of many such investigations are clear and unambiguous: While Q and W depend on the path, $\Delta U = Q - W$ is *independent of path*. The change in internal energy of a system during any thermodynamic process depends only on the initial and final states, not on the path leading from one to the other.

Figure 19.9 In a thermodynamic process, the internal energy U of a system may (a) increase ($\Delta U > 0$), (b) decrease ($\Delta U < 0$), or (c) remain the same ($\Delta U = 0$).

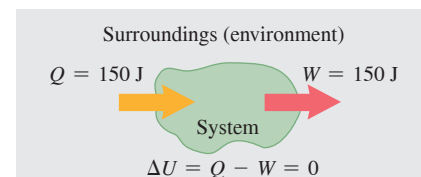
(a) More heat is added to system than system does work: Internal energy of system increases.



(b) More heat flows out of system than work is done: Internal energy of system decreases.



(c) Heat added to system equals work done by system: Internal energy of system unchanged.



BIO APPLICATION The First Law of Exercise Thermodynamics Your body is a thermodynamic system. When you exercise, your body does work (such as the work done to lift your body as a whole in a push-up). Hence $W > 0$. Your body also warms up during exercise; by perspiration and other means the body rids itself of this heat, so $Q < 0$. Since Q is negative and W is positive, $\Delta U = Q - W < 0$ and the body's internal energy decreases. That's why exercise helps you lose weight: It uses up some of the internal energy stored in your body in the form of fat.



Figure 19.10 The internal energy of a cup of coffee depends on just its thermodynamic state—how much water and ground coffee it contains, and what its temperature is. It does not depend on the history of how the coffee was prepared—that is, the thermodynamic path that led to its current state.



Experiment, then, is the ultimate justification for believing that a thermodynamic system in a specific state has a unique internal energy that depends only on that state. An equivalent statement is that the internal energy U of a system is a function of the state coordinates p , V , and T (actually, any two of these, since the three variables are related by the equation of state).

To say that the first law of thermodynamics, given by Eq. (19.4) or (19.5), represents conservation of energy for thermodynamic processes is correct, as far as it goes. But an important *additional* aspect of the first law is the fact that internal energy depends only on the state of a system (**Fig. 19.10**). In changes of state, the change in internal energy is independent of the path.

All this may seem a little abstract if you are satisfied to think of internal energy as microscopic mechanical energy. There's nothing wrong with that view, and we'll make use of it at various times during our discussion. But as for heat, a precise *operational* definition of internal energy must be independent of the detailed microscopic structure of the material.

Cyclic Processes and Isolated Systems

Two special cases of the first law of thermodynamics are worth mentioning. A process that eventually returns a system to its initial state is called a *cyclic process*. For such a process, the final state is the same as the initial state, and so the *total* internal energy change must be zero. Then

$$U_2 = U_1 \quad \text{and} \quad Q = W$$

If a net quantity of work W is done by the system during this process, an equal amount of energy must have flowed into the system as heat Q . But there is no reason either Q or W individually has to be zero (**Fig. 19.11**).

Another special case occurs in an *isolated system*, one that does no work on its surroundings and has no heat flow to or from its surroundings. For any process taking place in an isolated system,

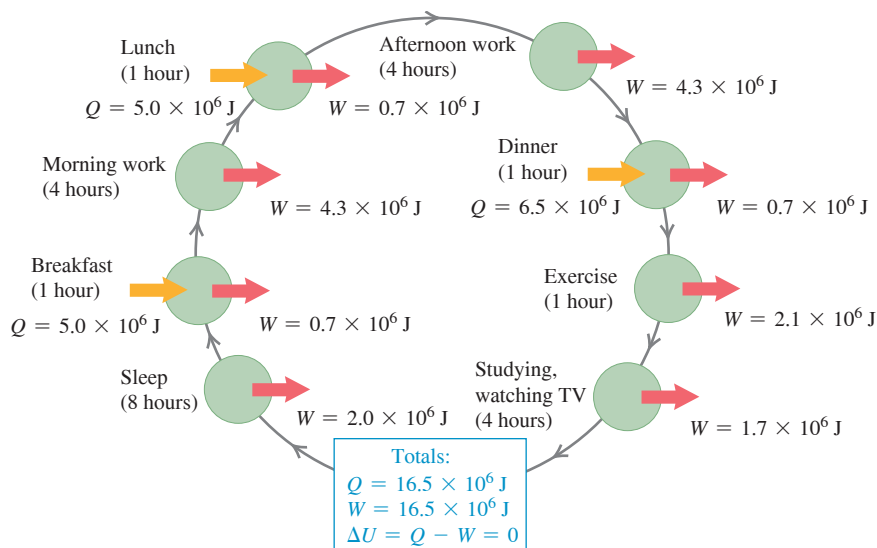
$$W = Q = 0$$

and therefore

$$U_2 = U_1 = \Delta U = 0$$

In other words, *the internal energy of an isolated system is constant*.

Figure 19.11 Every day, your body (a thermodynamic system) goes through a cyclic thermodynamic process like this one. Heat Q is added by metabolizing food, and your body does work W in breathing, walking, and other activities. If you return to the same state at the end of the day, $Q = W$ and the net change in your internal energy is zero.



PROBLEM-SOLVING STRATEGY 19.1 The First Law of Thermodynamics

IDENTIFY *the relevant concepts:* The first law of thermodynamics is the statement of the law of conservation of energy in its most general form. You can apply it to *any* thermodynamic process in which the internal energy of a system changes, heat flows into or out of the system, and/or work is done by or on the system.

SET UP *the problem* using the following steps:

1. Define the thermodynamic system to be considered.
2. If the thermodynamic process has more than one step, identify the initial and final states for each step.
3. List the known and unknown quantities and identify the target variables.
4. Confirm that you have enough equations. You can apply the first law, $\Delta U = Q - W$, just once to each step in a thermodynamic process, so you'll often need additional equations. These may include Eq. (19.2), $W = \int_{V_i}^{V_f} p dV$, which gives the work W done in a volume change, and the equation of state of the material that makes up the thermodynamic system (for an ideal gas, $pV = nRT$).

EXECUTE *the solution* as follows:

1. Be sure to use consistent units. If p is in Pa and V in m^3 , then W is in joules. If a heat capacity is given in terms of calories, convert it

to joules. When you use $n = m_{\text{total}}/M$ to relate total mass m_{total} to number of moles n , remember that if m_{total} is in kilograms, M must be in *kilograms* per mole; M is usually tabulated in *grams* per mole.

2. The internal energy change ΔU in any thermodynamic process or series of processes is independent of the path, whether the substance is an ideal gas or not. If you can calculate ΔU for *any* path between given initial and final states, you know ΔU for *every possible path* between those states; you can then relate the various energy quantities for any of those other paths.
3. In a process comprising several steps, tabulate Q , W , and ΔU for each step, with one line per step and with the Q 's, W 's, and ΔU 's forming columns (see Example 19.4). You can apply the first law to each line, and you can add each column and apply the first law to the sums. Do you see why?
4. Using steps 1–3, solve for the target variables.

EVALUATE *your answer:* Check your results for reasonableness. Ensure that each of your answers has the correct algebraic sign. A positive Q means that heat flows *into* the system; a negative Q means that heat flows *out of* the system. A positive W means that work is done *by* the system on its environment; a negative W means that work is done *on* the system by its environment.

EXAMPLE 19.2 Working off your dessert

WITH VARIATION PROBLEMS

You propose to climb several flights of stairs to work off the energy you took in by eating a 900-calorie hot fudge sundae. How high must you climb? Assume that your mass is 60.0 kg.

IDENTIFY and SET UP The thermodynamic system is your body. You climb the stairs to make the final state of the system the same as the initial state (no fatter, no leaner). There is therefore no net change in internal energy: $\Delta U = 0$. Eating the hot fudge sundae corresponds to a heat flow into your body, and you do work climbing the stairs. We can relate these quantities by using the first law of thermodynamics. We are given that $Q = 900$ food calories (900 kcal) of heat flow into your body. The work you must do to raise your mass m a height h is $W = mgh$; our target variable is h .

EXECUTE From the first law of thermodynamics, $\Delta U = 0 = Q - W$, so $W = mgh = Q$. Hence you must climb to height $h = Q/mg$. First convert units: $Q = (900 \text{ kcal})(4186 \text{ J/1 kcal}) = 3.77 \times 10^6 \text{ J}$. Then

$$h = \frac{Q}{mg} = \frac{3.77 \times 10^6 \text{ J}}{(60.0 \text{ kg})(9.80 \text{ m/s}^2)} = 6410 \text{ m}$$

EVALUATE We have unrealistically assumed 100% efficiency in the conversion of food energy into mechanical work. The actual efficiency is roughly 25%, so the work W you do as you “burn off” the sundae is only about $(0.25)(900 \text{ kcal}) = 225 \text{ kcal}$. (The remaining 75%, or 675 kcal, is transferred to your surroundings as heat.) Hence you actually must climb about $(0.25)(6410 \text{ m}) = 1600 \text{ m}$, or one *mile*! Do you really want that sundae?

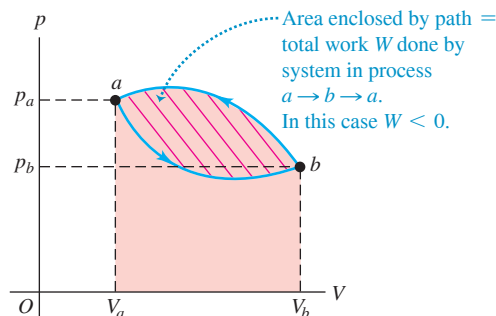
KEYCONCEPT The first law of thermodynamics states that the change in internal energy ΔU of a system equals the heat flow Q into the system minus the work W that the system does on its surroundings. Q is positive if heat flows into the system, negative if heat flows out; W is positive if the system does work on its surroundings, negative if the surroundings do work on the system.

EXAMPLE 19.3 A cyclic process

WITH VARIATION PROBLEMS

Figure 19.12 shows a pV -diagram for a *cyclic* process in which the initial and final states of some thermodynamic system are the same. The state of the system starts at point a and proceeds counterclockwise in the pV -diagram to point b , then back to a ; the total work is $W = -500 \text{ J}$. (a) Why is the work negative? (b) Find the change in internal energy and the heat added during this process.

Figure 19.12 The net work done by the system in process $a \rightarrow b \rightarrow a$ is -500 J . What would it have been if the process had proceeded clockwise in this pV -diagram?



Continued

IDENTIFY and SET UP We must relate the change in internal energy, the heat added, and the work done in a thermodynamic process. Hence we can apply the first law of thermodynamics. The process is cyclic, and it has two steps: $a \rightarrow b$ via the lower curve in Fig. 19.12 and $b \rightarrow a$ via the upper curve. We are asked only about the *entire* cyclic process $a \rightarrow b \rightarrow a$.

EXECUTE (a) The work done in any step equals the area under the curve in the pV -diagram, with the area taken as positive if $V_2 > V_1$ and negative if $V_2 < V_1$; this rule yields the signs that result from the actual integrations in Eq. (19.2), $W = \int_{V_1}^{V_2} p dV$. The area under the lower curve $a \rightarrow b$ is therefore positive, but it is smaller than the absolute value of the (negative) area under the upper curve $b \rightarrow a$. Therefore the net area (the area enclosed by the path, shown with red stripes) and the net work

W are negative. In other words, 500 J more work is done *on* the system than *by* the system in the complete process.

(b) In any cyclic process, $\Delta U = 0$, so $Q = W$. Here, that means $Q = -500$ J; that is, 500 J of heat flows *out* of the system.

EVALUATE In cyclic processes, the total work is positive if the process goes clockwise around the pV -diagram representing the cycle, and negative if the process goes counterclockwise (as here).

KEYCONCEPT In a cyclic thermodynamic process a system returns to the same state it was in initially, so the net change ΔU in internal energy is zero. The net work W done by the system in a cyclic process equals the area enclosed by the path that the system follows on a pV -diagram. Since $\Delta U = 0$, the net heat flow Q into the system in a cyclic process is equal to W .

EXAMPLE 19.4 Comparing thermodynamic processes

WITH VARIATION PROBLEMS

The pV -diagram of **Fig. 19.13** shows a series of thermodynamic processes. In process ab , 150 J of heat is added to the system; in process bd , 600 J of heat is added. Find (a) the internal energy change in process ab ; (b) the internal energy change in process abd (shown in light blue); and (c) the total heat added in process acd (shown in dark blue).

IDENTIFY and SET UP In each process we use $\Delta U = Q - W$ to determine the desired quantity. We are given $Q_{ab} = +150$ J and $Q_{bd} = +600$ J (both values are positive because heat is *added* to the system). Our target variables are (a) ΔU_{ab} , (b) ΔU_{abd} , and (c) Q_{acd} .

EXECUTE (a) No volume change occurs during process ab , so the system does no work: $W_{ab} = 0$ and so $\Delta U_{ab} = Q_{ab} = 150$ J.

(b) Process bd is an expansion at constant pressure, so from Eq. (19.3),

$$\begin{aligned} W_{bd} &= p(V_2 - V_1) \\ &= (8.0 \times 10^4 \text{ Pa})(5.0 \times 10^{-3} \text{ m}^3 - 2.0 \times 10^{-3} \text{ m}^3) = 240 \text{ J} \end{aligned}$$

The total work for the two-step process abd is then

$$W_{abd} = W_{ab} + W_{bd} = 0 + 240 \text{ J} = 240 \text{ J}$$

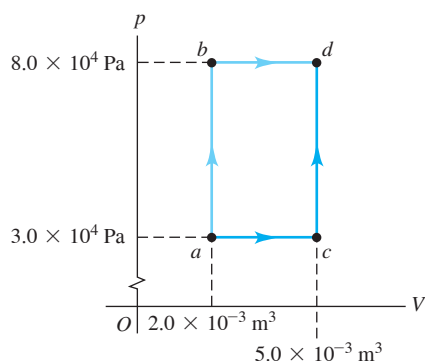
and the total heat is

$$Q_{abd} = Q_{ab} + Q_{bd} = 150 \text{ J} + 600 \text{ J} = 750 \text{ J}$$

Applying Eq. (19.4) to abd , we then have

$$\Delta U_{abd} = Q_{abd} - W_{abd} = 750 \text{ J} - 240 \text{ J} = 510 \text{ J}$$

Figure 19.13 A pV -diagram showing the various thermodynamic processes.



(c) Because ΔU is *independent of the path* from a to d , the internal energy change is the same for path acd as for path abd :

$$\Delta U_{acd} = \Delta U_{abd} = 510 \text{ J}$$

The total work for path acd is

$$\begin{aligned} W_{acd} &= W_{ac} + W_{cd} = p(V_2 - V_1) + 0 \\ &= (3.0 \times 10^4 \text{ Pa})(5.0 \times 10^{-3} \text{ m}^3 - 2.0 \times 10^{-3} \text{ m}^3) \\ &= 90 \text{ J} \end{aligned}$$

Now we apply Eq. (19.5) to process acd :

$$Q_{acd} = \Delta U_{acd} + W_{acd} = 510 \text{ J} + 90 \text{ J} = 600 \text{ J}$$

We tabulate the quantities above:

Step	Q	W	$\Delta U = Q - W$	Step	Q	W	$\Delta U = Q - W$
ab	150 J	0 J	150 J	ac	?	90 J	?
bd	600 J	240 J	360 J	cd	?	0 J	?
abd	750 J	240 J	510 J	acd	600 J	90 J	510 J

EVALUATE Be sure that you understand how each entry in the table above was determined. Although ΔU is the same (510 J) for abd and acd , W (240 J versus 90 J) and Q (750 J versus 600 J) are quite different. Although we couldn't find Q or ΔU for processes ac and cd , we could analyze the composite process acd by comparing it with process abd , which has the same initial and final states and for which we have more information.

KEYCONCEPT When a system starts in one state and ends in a different state, the internal energy change ΔU of the system is the same no matter what path the system takes between the two states. However, the heat flow Q into the system and the work W done by the system *do* depend on the path taken.

EXAMPLE 19.5 Thermodynamics of boiling water**WITH VARIATION PROBLEMS**

One gram of water (1 cm^3) becomes 1671 cm^3 of steam when boiled at a constant pressure of 1 atm ($1.013 \times 10^5 \text{ Pa}$). The heat of vaporization at this pressure is $L_v = 2.256 \times 10^6 \text{ J/kg}$. Compute (a) the work done by the water when it vaporizes and (b) its increase in internal energy.

IDENTIFY and SET UP The heat added causes the system (water) to change phase from liquid to vapor. We can analyze this process by using the first law of thermodynamics. The water is boiled at constant pressure, so we can use Eq. (19.3) to calculate the work W done by the vaporizing water as it expands. We are given the mass of water and the heat of vaporization, so we can use Eq. (17.20), $Q = mL_v$, to calculate the heat Q added to the water. We can then find the internal energy change from Eq. (19.4), $\Delta U = Q - W$.

EXECUTE (a) From Eq. (19.3), the water does work

$$W = p(V_2 - V_1) \\ = (1.013 \times 10^5 \text{ Pa})(1671 \times 10^{-6} \text{ m}^3 - 1 \times 10^{-6} \text{ m}^3) = 169 \text{ J}$$

(b) From Eq. (17.20), the heat added to the water is

$$Q = mL_v = (10^{-3} \text{ kg})(2.256 \times 10^6 \text{ J/kg}) = 2256 \text{ J}$$

Then from Eq. (19.4),

$$\Delta U = Q - W = 2256 \text{ J} - 169 \text{ J} = 2087 \text{ J}$$

EVALUATE To vaporize 1 g of water, we must add 2256 J of heat, most of which (2087 J) remains in the system as an increase in internal energy. The remaining 169 J leaves the system as the system expands from liquid to vapor and does work against the surroundings. (The increase in internal energy is associated mostly with the attractive intermolecular forces. The associated potential energies are greater after work has been done to pull apart the molecules in the liquid, forming the vapor state. It's like increasing gravitational potential energy by pulling an elevator farther from the center of the earth.)

KEYCONCEPT For a system that undergoes a thermodynamic process, if you can calculate any two of the quantities ΔU (internal energy change), Q (heat flow into the system), and W (work done by the system), you can calculate the third using the first law of thermodynamics, $\Delta U = Q - W$.

Infinitesimal Changes of State

In the preceding examples the initial and final states differ by a finite amount. Later we'll consider *infinitesimal* changes of state in which a small amount of heat dQ is added to the system, the system does a small amount of work dW , and its internal energy changes by an amount dU . For such a process,

$$\text{First law of thermodynamics, infinitesimal process:} \quad \text{Infinitesimal internal energy change} \quad dU = dQ - dW \quad (19.6)$$

Infinitesimal heat added Infinitesimal work done

For the systems we'll discuss, the work dW is given by $dW = p dV$, so we can also state the first law as

$$dU = dQ - p dV \quad (19.7)$$

TEST YOUR UNDERSTANDING OF SECTION 19.4 Rank the following thermodynamic processes according to the change in internal energy in each process, from most positive to most negative. (i) As you do 250 J of work on a system, it transfers 250 J of heat to its surroundings; (ii) as you do 250 J of work on a system, it absorbs 250 J of heat from its surroundings; (iii) as a system does 250 J of work on you, it transfers 250 J of heat to its surroundings; (iv) as a system does 250 J of work on you, it absorbs 250 J of heat from its surroundings.

ANSWER

ANSWER (i) $\Delta U = 0$; (ii) $\Delta U = 500 \text{ J}$; (iii) $\Delta U = -500 \text{ J}$; (iv) $\Delta U = 0$. In the expression $\Delta U = Q - W$, Q is the heat added to the system and W is the work done by the system. If heat is transferred from the system to its surroundings, Q is negative; if work is done on the system, W is negative. Hence we have (i) $\Delta U = 250 \text{ J} - 250 \text{ J} = 0$; (ii) $\Delta U = 250 \text{ J} - (-250 \text{ J}) = 500 \text{ J}$; (iii) $\Delta U = (-250 \text{ J}) - 250 \text{ J} = -500 \text{ J}$; and (iv) $\Delta U = 250 \text{ J} - 0 = 250 \text{ J}$.

19.5 KINDS OF THERMODYNAMIC PROCESSES

In this section we describe four specific kinds of thermodynamic processes that occur often in practical situations. We can summarize these briefly as “no heat transfer” or *adiabatic*, “constant volume” or *isochoric*, “constant pressure” or *isobaric*, and “constant temperature” or *isothermal*. For some of these processes we can use a simplified form of the first law of thermodynamics.

Figure 19.14 When the cork is popped on a bottle of champagne, the pressurized gases inside the bottle expand rapidly and do work on the outside air ($W > 0$). There is little time for the gases to exchange heat with their surroundings, so the expansion is nearly adiabatic ($Q = 0$). Hence the internal energy of the expanding gases decreases ($\Delta U = -W < 0$) and their temperature drops. This makes water vapor condense and form a miniature cloud.

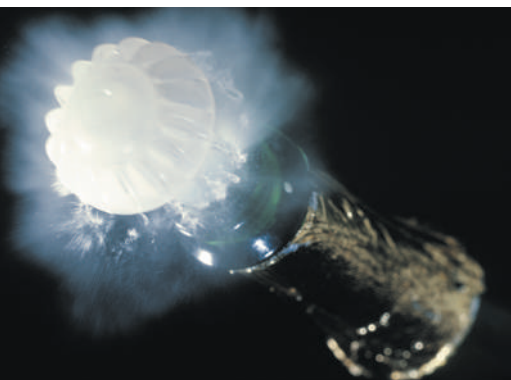


Figure 19.15 Most cooking involves isobaric processes. That's because the air pressure above a saucepan or frying pan, or inside a microwave oven, remains essentially constant while the food is being heated.



Adiabatic Process

An **adiabatic process** (pronounced “ay-dee-ah-bat-ic”) is defined as one with no heat transfer into or out of a system; $Q = 0$. We can prevent heat flow either by surrounding the system with thermally insulating material or by carrying out the process so quickly that there is not enough time for appreciable heat flow. From the first law we find that for every adiabatic process,

$$U_2 - U_1 = \Delta U = -W \quad (\text{adiabatic process}) \quad (19.8)$$

When a system expands adiabatically, W is positive (the system does work on its surroundings), so ΔU is negative and the internal energy decreases. When a system is *compressed* adiabatically, W is negative (work is done on the system by its surroundings) and U increases. In many (but not all) systems an increase of internal energy is accompanied by a rise in temperature, and a decrease in internal energy by a drop in temperature (Fig. 19.14).

The compression stroke in an internal-combustion engine is an approximately adiabatic process. The temperature rises as the air–fuel mixture in the cylinder is compressed. The expansion of the burned fuel during the power stroke is also an approximately adiabatic expansion with a drop in temperature. In Section 19.8 we’ll consider adiabatic processes in an ideal gas.

Isochoric Process

An **isochoric process** (pronounced “eye-so-kor-ic”) is a *constant-volume* process. When the volume of a thermodynamic system is constant, it does no work on its surroundings. Then $W = 0$ and

$$U_2 - U_1 = \Delta U = Q \quad (\text{isochoric process}) \quad (19.9)$$

In an isochoric process, all the energy added as heat remains in the system as an increase in internal energy. Heating a gas in a closed constant-volume container is an example of an isochoric process. The processes *ab* and *cd* in Example 19.4 are also examples of isochoric processes. (Note that there are types of work that do not involve a volume change. For example, we can do work on a fluid by stirring it. In some literature, “isochoric” is used to mean that no work of any kind is done.)

Isobaric Process

An **isobaric process** (pronounced “eye-so-bear-ic”) is a *constant-pressure* process. In general, none of the three quantities ΔU , Q , and W is zero in an isobaric process, but calculating W is easy nonetheless. From Eq. (19.3),

$$W = p(V_2 - V_1) \quad (\text{isobaric process}) \quad (19.10)$$

Boiling water at constant pressure is an isobaric process (Fig. 19.15).

Isothermal Process

An **isothermal process** is a *constant-temperature* process. For a process to be isothermal, any heat flow into or out of the system must occur slowly enough that thermal equilibrium is maintained. In general, none of the quantities ΔU , Q , or W is zero in an isothermal process.

In some special cases the internal energy of a system depends *only* on its temperature, not on its pressure or volume. The most familiar system having this special property is an ideal gas, as we’ll discuss in the next section. For such systems, if the temperature is constant, the internal energy is also constant; $\Delta U = 0$ and $Q = W$. That is, any energy entering the system as heat Q must leave it again as work W done by the system. Example 19.1, involving an ideal gas, is an example of an isothermal process in which U is also constant. For most systems other than ideal gases, the internal energy depends on pressure as well as temperature, so U may vary even when T is constant.

Figure 19.16 shows a pV -diagram for these four processes for a constant amount of an ideal gas. The path followed in an adiabatic process (a to 1) is called an **adiabat**. A vertical line (constant volume) is an **isochor**, a horizontal line (constant pressure) is an **isobar**, and a curve of constant temperature (shown as light blue lines in Fig. 19.16) is an **isotherm**.

TEST YOUR UNDERSTANDING OF SECTION 19.5 Which of the processes in Fig. 19.7 are isochoric? Which are isobaric? Is it possible to tell if any of the processes are isothermal or adiabatic?

ANSWER

I 1 $\leftarrow$ 4 and 3 $\rightarrow$ 2 are isochoric; 1 $\leftarrow$ 3 and 4 $\rightarrow$ 2 are isobaric; no 2 are isothermal; no In a pV -diagram like those shown in Fig. 19.7, isochoric processes are represented by vertical lines (lines of constant volume) and isobaric processes are represented by horizontal lines (lines of constant pressure). The process 1 $\rightarrow$ 2 in Fig. 19.7 is shown as a curved line, which superficially resembles the adiabatic and isothermal processes for an ideal gas in Fig. 19.16. Without more information we can't tell whether process 1 $\rightarrow$ 2 is isothermal, adiabatic, or neither.

19.6 INTERNAL ENERGY OF AN IDEAL GAS

We now show that for an ideal gas, the internal energy U depends only on temperature, not on pressure or volume. Let's think again about the free-expansion experiment described in Section 19.3. A thermally insulated container with rigid walls is divided into two compartments by a partition (**Fig. 19.17**). One compartment has a quantity of an ideal gas and the other is evacuated.

When the partition is removed or broken, the gas expands to fill both parts of the container. There is no heat flow through the insulation, and the gas does no work on its surroundings because the walls of the container don't move. So both Q and W are zero and the internal energy U is constant.

Does the *temperature* T change during a free expansion? Suppose it *does* change, while the internal energy stays the same. In that case we have to conclude that the internal energy depends on both T and the volume V or on both T and the pressure p , but certainly not on T alone. But if T is constant during a free expansion, for which we know that U is constant even though both p and V change, then we have to conclude that U depends only on T , not on p or V .

Many experiments have shown that when a low-density gas (essentially an ideal gas) undergoes a free expansion, its temperature *does not* change. The conclusion is:

The internal energy U of an ideal gas depends only on its temperature T , not on its pressure or volume.

This property, in addition to the ideal-gas equation of state, is part of the ideal-gas model. We'll make frequent use of this property.

For nonideal gases, some temperature change occurs during free expansions, even though the internal energy is constant. This shows that the internal energy cannot depend *only* on temperature; it must depend on pressure as well. From the microscopic viewpoint, in which internal energy U is the sum of the kinetic and potential energies for all the particles that make up the system, this is not surprising. Nonideal gases usually have attractive intermolecular forces, and when molecules move farther apart, the associated potential energies increase. If the total internal energy is constant, the kinetic energies must decrease. Temperature is directly related to molecular *kinetic* energy, and for such a gas a free expansion is usually accompanied by a *drop* in temperature.

TEST YOUR UNDERSTANDING OF SECTION 19.6 Is the internal energy of a solid likely to be independent of its volume, as is the case for an ideal gas? Explain your reasoning. (*Hint:* See Fig. 18.20.)

ANSWER **I** no Using the model of a solid in Fig. 18.20, we can see that the internal energy of a solid *does* depend on its volume. Compressing the solid means compressing the "springs" between the atoms, thereby increasing their stored potential energy and hence the internal energy of the solid.

Figure 19.16 Four different processes for a constant amount of an ideal gas, all starting at state a . For the adiabatic process, $Q = 0$; for the isochoric process, $W = 0$; and for the isothermal process, $\Delta U = 0$. The temperature increases only during the isobaric expansion.

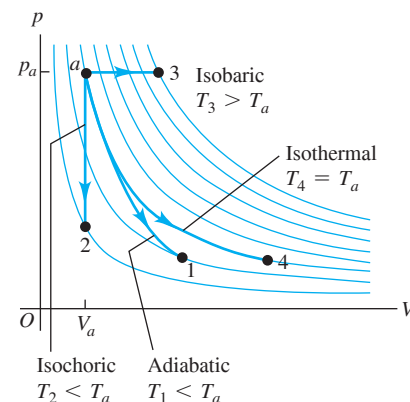


Figure 19.17 The partition is broken (or removed) to start the free expansion of gas into the vacuum region.

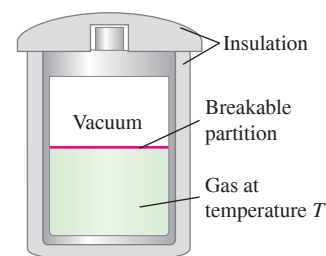
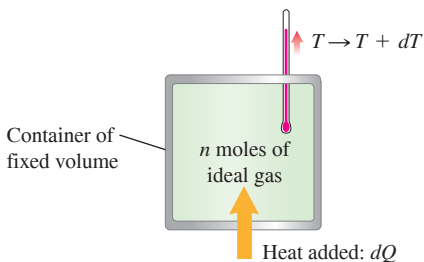


Figure 19.18 Measuring the molar heat capacity of an ideal gas (a) at constant volume and (b) at constant pressure.

(a) Constant volume: $dQ = nC_V dT$



(b) Constant pressure: $dQ = nC_p dT$

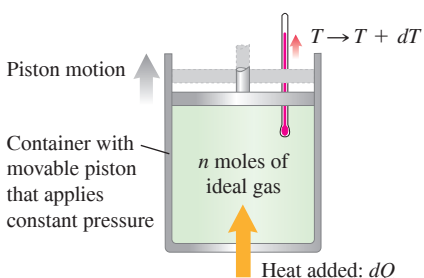
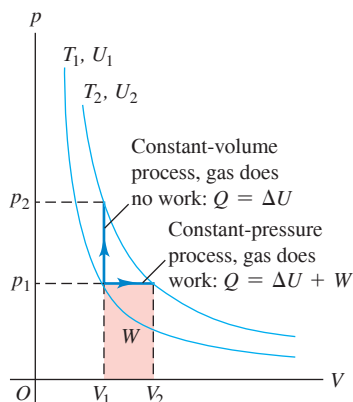


Figure 19.19 Raising the temperature of an ideal gas from T_1 to T_2 by a constant-volume or a constant-pressure process. For an ideal gas, U depends only on T , so ΔU is the same for both processes. But for the constant-pressure process, more heat Q must be added to both increase U and do work W . Hence $C_p > C_V$.



19.7 HEAT CAPACITIES OF AN IDEAL GAS

We defined specific heat and molar heat capacity in Section 17.5. We also remarked at the end of that section that the specific heat or molar heat capacity of a substance depends on the conditions under which the heat is added. The heat capacity of a gas is usually measured in a closed container under constant-volume conditions. The corresponding heat capacity is the **molar heat capacity at constant volume**, denoted by C_V . Heat capacity measurements for solids and liquids are usually carried out under constant atmospheric pressure, and we call the corresponding heat capacity the **molar heat capacity at constant pressure**, C_p .

Let's consider C_V and C_p for an ideal gas. To measure C_V , we raise the temperature of an ideal gas in a rigid container with constant volume, ignoring its thermal expansion (Fig. 19.18a). To measure C_p , we let the gas expand just enough to keep the pressure constant as the temperature rises (Fig. 19.18b).

Why should these two molar heat capacities be different? The answer lies in the first law of thermodynamics. In a constant-volume temperature increase, the system does no work, and the change in internal energy ΔU equals the heat added Q . In a constant-pressure temperature increase, on the other hand, the volume *must* increase; otherwise, the pressure (given by the ideal-gas equation of state, $p = nRT/V$) could not remain constant. As the material expands, it does an amount of work W . According to the first law,

$$Q = \Delta U + W \quad (19.11)$$

For a given temperature increase, the internal energy change ΔU of an ideal gas has the same value no matter what the process (remember that the internal energy of an ideal gas depends only on temperature, not on pressure or volume). Equation (19.11) then shows that the heat input for a constant-pressure process must be *greater* than that for a constant-volume process because additional energy must be supplied to account for the work W done during the expansion. So C_p is greater than C_V for an ideal gas. The pV -diagram in Fig. 19.19 shows this relationship. For air, C_p is 40% greater than C_V .

For a very few substances (one of which is water between 0°C and 4°C) the volume *decreases* during heating. In this case, W is negative and the internal energy change ΔU is greater than the heat input Q .

Relating C_p and C_V for an Ideal Gas

We can derive a simple relationship between C_p and C_V for an ideal gas. First consider the constant-volume process. We place n moles of an ideal gas at temperature T in a constant-volume container. We place it in thermal contact with a hotter object; an infinitesimal quantity of heat dQ flows into the gas, and its temperature increases by an infinitesimal amount dT . By the definition of C_V , the molar heat capacity at constant volume,

$$dQ = nC_V dT \quad (19.12)$$

The pressure increases during this process, but the gas does no work ($dW = 0$) because the volume is constant. The first law in differential form, Eq. (19.6), is $dQ = dU + dW$. Since $dW = 0$, $dQ = dU$ and Eq. (19.12) can also be written as

$$dU = nC_V dT \quad (19.13)$$

Now consider a constant-pressure process with the same temperature change dT . We place the same gas in a cylinder with a piston that we allow to move just enough to maintain constant pressure (Fig. 19.18b). Again we bring the system into contact with a hotter object. As heat flows into the gas, it expands at constant pressure and does work. By the definition of C_p , the molar heat capacity at constant pressure, the amount of heat entering the gas is

$$dQ = nC_p dT \quad (19.14)$$

The work dW done by the gas in this constant-pressure process is

$$dW = p dV$$

We can also express dW in terms of the temperature change dT by using the ideal-gas equation of state, $pV = nRT$. Because p is constant, the change in V is proportional to the change in T :

$$dW = p dV = nR dT \quad (19.15)$$

Now substitute Eqs. (19.14) and (19.15) into the first law, $dQ = dU + dW$:

$$nC_p dT = dU + nR dT \quad (19.16)$$

Here comes the crux of the calculation. The internal energy change dU for the constant-pressure process is again given by Eq. (19.13), $dU = nC_V dT$, *even though now the volume is not constant*. Why is this so? From Section 19.6, one of the special properties of an ideal gas is that its internal energy depends *only* on temperature. Thus the *change* in internal energy during any process must be determined only by the temperature change. If Eq. (19.13) is valid for an ideal gas for one particular kind of process, it must be valid for an ideal gas for *every* kind of process with the same dT . So we may replace dU in Eq. (19.16) by $nC_V dT$:

$$nC_p dT = nC_V dT + nR dT$$

When we divide each term by the common factor $n dT$, we get

$$\text{For an ideal gas:} \quad C_p = C_V + R \quad (19.17)$$

Molar heat capacity at constant pressure
Gas constant
Molar heat capacity at constant volume

As we predicted, the molar heat capacity of an ideal gas at constant pressure is *greater* than the molar heat capacity at constant volume; the difference is the gas constant R .

We have used the ideal-gas model to derive Eq. (19.17), but it turns out to be obeyed to within a few percent by many real gases at moderate pressures. **Table 19.1** gives measured values of C_p and C_V for several real gases at low pressures; the difference in most cases is approximately $R = 8.314 \text{ J/mol} \cdot \text{K}$.

The table also shows that the molar heat capacity of a gas is related to its molecular structure, as we discussed in Section 18.4. In fact, the first three columns of Table 19.1 are the same as Table 18.1.

The Ratio of Heat Capacities

The last column of Table 19.1 lists the values of the dimensionless **ratio of heat capacities**, C_p/C_V , denoted by γ (the Greek letter gamma):

$$\text{Ratio of heat capacities} \quad \gamma = \frac{C_p}{C_V} \quad (19.18)$$

Molar heat capacity at constant pressure
Molar heat capacity at constant volume

TABLE 19.1 Molar Heat Capacities of Gases at Low Pressure

Type of Gas	Gas	C_V (J/mol · K)	C_p (J/mol · K)	$C_p - C_V$ (J/mol · K)	$\gamma = C_p/C_V$
Monatomic	He	12.47	20.78	8.31	1.67
	Ar	12.47	20.78	8.31	1.67
Diatomic	H ₂	20.42	28.74	8.32	1.41
	N ₂	20.76	29.07	8.31	1.40
	O ₂	20.85	29.17	8.32	1.40
	CO	20.85	29.16	8.31	1.40
Polyatomic	CO ₂	28.46	36.94	8.48	1.30
	SO ₂	31.39	40.37	8.98	1.29
	H ₂ S	25.95	34.60	8.65	1.33

(This is sometimes called the “ratio of specific heats.”) For gases, C_p is always greater than C_V and γ is always greater than unity. We’ll see in the next section that γ plays an important role in adiabatic processes for an ideal gas.

We can use our kinetic-theory discussion of the molar heat capacity of an ideal gas (see Section 18.4) to predict values of γ . As an example, an ideal monatomic gas has $C_V = \frac{3}{2}R$. From Eq. (19.17),

$$C_p = C_V + R = \frac{3}{2}R + R = \frac{5}{2}R$$

so

$$\gamma = \frac{C_p}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3} = 1.67$$

As Table 19.1 shows, this agrees well with values of γ computed from measured heat capacities. For most diatomic gases near room temperature, $C_V = \frac{5}{2}R$, $C_p = C_V + R = \frac{7}{2}R$, and

$$\gamma = \frac{C_p}{C_V} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = \frac{7}{5} = 1.40$$

also in good agreement with measured values.

Here’s a final reminder: For an ideal gas, the internal energy change in *any* process is given by $\Delta U = nC_V \Delta T$, *whether the volume is constant or not*. This relationship holds for other substances *only* when the volume is constant.

CAUTION A few hints about ΔU , Q , and W for an ideal gas Remember that if the gas temperature T increases, the internal energy change ΔU of the gas is positive; if T decreases, ΔU is negative. Remember, too, that the work W done by the gas is positive if the gas expands and negative if the gas is compressed. Finally, you can calculate the heat flow Q into the gas directly only in certain special processes (isochoric, isobaric, or adiabatic processes); for other processes, you must first calculate ΔU and W and then find Q using the first law of thermodynamics. **|**

EXAMPLE 19.6 Cooling your room

WITH VARIATION PROBLEMS

A typical dorm room or bedroom contains about 2500 moles of air. Find the change in the internal energy of this much air when it is cooled from 35.0°C to 26.0°C at a constant pressure of 1.00 atm. Treat the air as an ideal gas with $\gamma = 1.400$.

IDENTIFY and SET UP Our target variable is the change in the internal energy ΔU of an ideal gas in a constant-pressure process. We are given the number of moles, the temperature change, and the value of γ for air. We use Eq. (19.13), $\Delta U = nC_V \Delta T$, which gives the internal energy change for an ideal gas in *any* process, *whether the volume is constant or not*. [See the discussion following Eq. (19.16).] We use Eqs. (19.17) and (19.18) to find C_V .

EXECUTE From Eqs. (19.17) and (19.18),

$$\gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V}$$

$$C_V = \frac{R}{\gamma - 1} = \frac{8.314 \text{ J/mol} \cdot \text{K}}{1.400 - 1} = 20.79 \text{ J/mol} \cdot \text{K}$$

Then from Eq. (19.13),

$$\begin{aligned}\Delta U &= nC_V \Delta T \\ &= (2500 \text{ mol})(20.79 \text{ J/mol} \cdot \text{K})(26.0^\circ\text{C} - 35.0^\circ\text{C}) \\ &= -4.68 \times 10^5 \text{ J}\end{aligned}$$

EVALUATE To cool 2500 moles of air from 35.0°C to 26.0°C, a room air conditioner must extract this much internal energy from the air and transfer it to the air outside. In Chapter 20 we’ll discuss how this is done.

KEYCONCEPT When n moles of an ideal gas undergo a thermodynamic process in which the temperature changes by ΔT , the change in internal energy is equal to the product of n , ΔT , and the molar heat capacity at constant volume C_V : $\Delta U = nC_V \Delta T$. This is true whether or not the volume of the system remains constant.

TEST YOUR UNDERSTANDING OF SECTION 19.7 You want to cool a storage cylinder containing 10 moles of compressed gas from 30°C to 20°C. For which kind of gas would this be easiest? (i) A monatomic gas; (ii) a diatomic gas; (iii) a polyatomic gas; (iv) it would be equally easy for all of these.

ANSWER

(i) For a given number of moles n and a given temperature change ΔT , the amount of heat that must be transferred out of a fixed volume of air is $Q = nC_V \Delta T$. Hence the amount of heat transfer required is least for the gas with the smallest value of C_V . From Table 19.1, C_V is smallest for monatomic gases.

19.8 ADIABATIC PROCESSES FOR AN IDEAL GAS

An adiabatic process, defined in Section 19.5, is a process in which no heat transfer takes place between a system and its surroundings. Zero heat transfer is an idealization, but a process is approximately adiabatic if the system is well insulated or if the process takes place so quickly that there is not enough time for appreciable heat flow to occur.

In an adiabatic process, $Q = 0$, so from the first law, $\Delta U = -W$. An adiabatic process for an ideal gas is shown in the pV -diagram of **Fig. 19.20**. As the gas expands from volume V_a to V_b , it does positive work, so its internal energy decreases and its temperature drops. If point a , representing the initial state, lies on an isotherm at temperature $T + dT$, then point b for the final state is on a different isotherm at a lower temperature T . For an adiabatic *compression* from V_b to V_a the situation is reversed and the temperature rises.

The air in the output hoses of air compressors used to inflate tires and to fill scuba tanks is always warmer than the air entering the compressor; this is because the compression is rapid and hence approximately adiabatic. Adiabatic *cooling* occurs when you open a bottle of your favorite carbonated beverage. The gas just above the beverage surface expands rapidly in a nearly adiabatic process; the gas temperature drops so much that water vapor in the gas condenses, forming a miniature cloud (see Fig. 19.14).

CAUTION “Heating” and “cooling” without heat When we talk about “adiabatic heating” and “adiabatic cooling,” we really mean “raising the temperature” and “lowering the temperature,” respectively. In an adiabatic process, the temperature change is due to work done by or on the system; there is *no* heat flow at all. ■

Adiabatic Ideal Gas: Relating V , T , and p

We can derive a relationship between volume and temperature changes for an infinitesimal adiabatic process in an ideal gas. Equation (19.13) gives the internal energy change dU for *any* process for an ideal gas, adiabatic or not, so we have $dU = nC_V dT$. Also, the work done by the gas during the process is given by $dW = p dV$. Then, since $dU = -dW$ for an adiabatic process, we have

$$nC_V dT = -p dV \quad (19.19)$$

To obtain a relationship containing only the volume V and temperature T , we eliminate p by using the ideal-gas equation in the form $p = nRT/V$. Substituting this into Eq. (19.19) and rearranging, we get

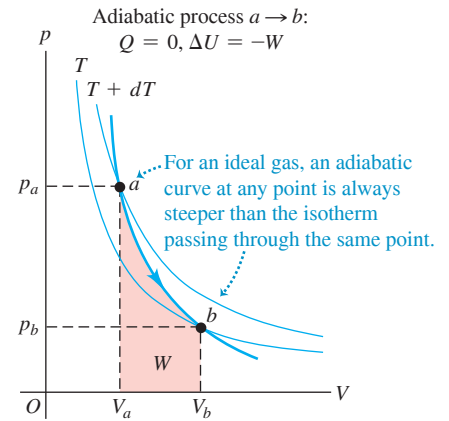
$$\begin{aligned} nC_V dT &= -\frac{nRT}{V} dV \\ \frac{dT}{T} + \frac{R}{C_V} \frac{dV}{V} &= 0 \end{aligned}$$

The coefficient R/C_V can be expressed in terms of $\gamma = C_p/C_V$. We have

$$\begin{aligned} \frac{R}{C_V} &= \frac{C_p - C_V}{C_V} = \frac{C_p}{C_V} - 1 = \gamma - 1 \\ \frac{dT}{T} + (\gamma - 1) \frac{dV}{V} &= 0 \end{aligned} \quad (19.20)$$

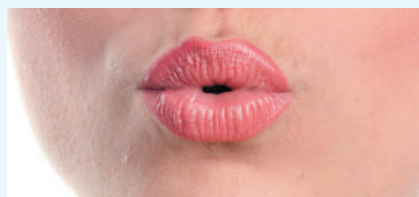
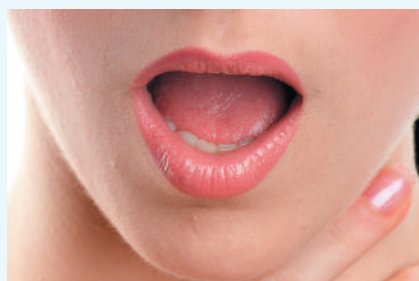
Because γ is always greater than unity for a gas, $(\gamma - 1)$ is always positive. This means that in Eq. (19.20), dV and dT always have opposite signs. An adiabatic *expansion* of an ideal gas ($dV > 0$) always occurs with a *drop* in temperature ($dT < 0$), and an adiabatic *compression* ($dV < 0$) always occurs with a *rise* in temperature ($dT > 0$); this confirms our earlier prediction.

Figure 19.20 A pV -diagram of an adiabatic ($Q = 0$) process for an ideal gas. As the gas expands from V_a to V_b , it does positive work W on its environment, its internal energy decreases ($\Delta U = -W < 0$), and its temperature drops from $T + dT$ to T . (An adiabatic process is also shown in Fig. 19.16.)



BIO APPLICATION Exhaling

Adiabatically Put your hand a few centimeters in front of your mouth, open your mouth wide, and exhale. Your breath will feel warm on your hand, because the exhaled gases emerge at roughly the temperature of your body's interior. Now purse your lips as though you were going to whistle, and again blow on your hand. The exhaled gases will feel much cooler. In this case the gases undergo a rapid, essentially adiabatic expansion as they emerge from between your lips, so the temperature of the exhaled gases decreases.



For finite changes in temperature and volume we integrate Eq. (19.20), obtaining

$$\ln T + (\gamma - 1) \ln V = \text{constant}$$

$$\ln T + \ln V^{\gamma-1} = \text{constant}$$

$$\ln(TV^{\gamma-1}) = \text{constant}$$

and finally,

$$TV^{\gamma-1} = \text{constant} \quad (19.21)$$

Thus for an initial state (T_1, V_1) and a final state (T_2, V_2) ,

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \quad (\text{adiabatic process, ideal gas}) \quad (19.22)$$

Because we have used the ideal-gas equation in our derivation of Eqs. (19.21) and (19.22), the T 's must always be *absolute* (Kelvin) temperatures.

We can also convert Eq. (19.21) into a relationship between pressure and volume by eliminating T , using the ideal-gas equation in the form $T = pV/nR$. Substituting this into Eq. (19.21), we find

$$\frac{pV}{nR} V^{\gamma-1} = \text{constant}$$

or, because n and R are constant,

$$pV^\gamma = \text{constant} \quad (19.23)$$

For an initial state (p_1, V_1) and a final state (p_2, V_2) , Eq. (19.23) becomes

$$p_1 V_1^\gamma = p_2 V_2^\gamma \quad (\text{adiabatic process, ideal gas}) \quad (19.24)$$

We can also calculate the *work* done by an ideal gas during an adiabatic process. We know that $Q = 0$ and $W = -\Delta U$ for *any* adiabatic process. For an ideal gas, $\Delta U = nC_V(T_2 - T_1)$. If the number of moles n and the initial and final temperatures T_1 and T_2 are known, we have simply

Work done by an ideal gas, adiabatic process

$$W = nC_V(T_1 - T_2) \quad (19.25)$$

Number of moles

Initial temperature

Final temperature

Molar heat capacity at constant volume

We may also use $pV = nRT$ in this equation to obtain

Work done by an ideal gas, adiabatic process

Molar heat capacity at constant volume

Initial pressure, volume

Gas constant

Ratio of heat capacities

Final pressure, volume

$$W = \frac{C_V}{R} (p_1 V_1 - p_2 V_2) = \frac{1}{\gamma - 1} (p_1 V_1 - p_2 V_2) \quad (19.26)$$

(We used the result $C_V = R/(\gamma - 1)$ from Example 19.6.) If the process is an expansion, the temperature drops, T_1 is greater than T_2 , $p_1 V_1$ is greater than $p_2 V_2$, and the work is *positive*. If the process is a compression, the work is *negative*.

Throughout this analysis of adiabatic processes we have used the ideal-gas equation of state, which is valid only for *equilibrium* states. Strictly speaking, our results are valid only for a process that is fast enough to prevent appreciable heat exchange with the surroundings (so that $Q = 0$ and the process is adiabatic), yet slow enough that the system does not depart very much from thermal and mechanical equilibrium. Even when these conditions are not strictly satisfied, though, Eqs. (19.22), (19.24), and (19.26) give useful approximate results.

EXAMPLE 19.7 Adiabatic compression in a diesel engine**WITH VARIATION PROBLEMS**

The compression ratio of a diesel engine is 15.0 to 1; that is, air in a cylinder is compressed to $\frac{1}{(15.0)}$ of its initial volume (**Fig. 19.21**). (a) If the initial pressure is $1.01 \times 10^5 \text{ Pa}$ and the initial temperature is 27°C (300 K), find the final pressure and the temperature after adiabatic compression. (b) How much work does the gas do during the compression if the initial volume of the cylinder is $1.00 \text{ L} = 1.00 \times 10^{-3} \text{ m}^3$? Use the values $C_V = 20.8 \text{ J/mol} \cdot \text{K}$ and $\gamma = 1.400$ for air.

IDENTIFY and SET UP This problem involves the adiabatic compression of an ideal gas, so we can use the ideas of this section. In part (a) we are given the initial pressure and temperature $p_1 = 1.01 \times 10^5 \text{ Pa}$ and $T_1 = 300 \text{ K}$; the ratio of initial and final volumes is $V_1/V_2 = 15.0$. We use Eq. (19.22) to find the final temperature T_2 and Eq. (19.24) to find the final pressure p_2 . In part (b) our target variable is W , the work done by the gas during the adiabatic compression. We use Eq. (19.26) to calculate W .

EXECUTE (a) From Eqs. (19.22) and (19.24),

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = (300 \text{ K})(15.0)^{0.40} = 886 \text{ K} = 613^\circ\text{C}$$

$$\begin{aligned} p_2 &= p_1 \left(\frac{V_1}{V_2} \right)^\gamma = (1.01 \times 10^5 \text{ Pa})(15.0)^{1.40} \\ &= 44.8 \times 10^5 \text{ Pa} = 44 \text{ atm} \end{aligned}$$

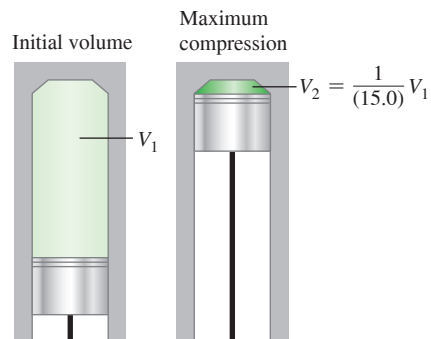
(b) From Eq. (19.26), the work done is

$$W = \frac{1}{\gamma - 1} (p_1 V_1 - p_2 V_2)$$

Using $V_1/V_2 = 15.0$, we have

$$\begin{aligned} W &= \frac{1}{1.400 - 1} \left[(1.01 \times 10^5 \text{ Pa})(1.00 \times 10^{-3} \text{ m}^3) \right. \\ &\quad \left. - (44.8 \times 10^5 \text{ Pa}) \left(\frac{1.00 \times 10^{-3} \text{ m}^3}{15.0} \right) \right] \\ &= -494 \text{ J} \end{aligned}$$

Figure 19.21 Adiabatic compression of air in a cylinder of a diesel engine.



EVALUATE If the compression had been isothermal, the final pressure would have been 15.0 atm. Because the temperature also increases during an adiabatic compression, the final pressure is much greater. When fuel is injected into the cylinders near the end of the compression stroke, the high temperature of the air attained during compression causes the fuel to ignite spontaneously without the need for spark plugs.

We can check our result in part (b) by using Eq. (19.25). The number of moles of gas in the cylinder is

$$n = \frac{p_1 V_1}{RT_1} = \frac{(1.01 \times 10^5 \text{ Pa})(1.00 \times 10^{-3} \text{ m}^3)}{(8.314 \text{ J/mol} \cdot \text{K})(300 \text{ K})} = 0.0405 \text{ mol}$$

Then Eq. (19.25) gives

$$\begin{aligned} W &= nC_V(T_1 - T_2) \\ &= (0.0405 \text{ mol})(20.8 \text{ J/mol} \cdot \text{K})(300 \text{ K} - 886 \text{ K}) = -494 \text{ J} \end{aligned}$$

The work is negative because the gas is compressed.

KEYCONCEPT In an adiabatic process there is no heat flow into or out of the system, so $Q = 0$ and the internal energy equals the negative of the work done by the system: $\Delta U = -W$. If the system is an ideal gas, the temperature T , volume V , and pressure p all change in the process, but the quantities $TV^{\gamma-1}$ and PV^γ remain constant (γ is the ratio of specific heats for the gas).

TEST YOUR UNDERSTANDING OF SECTION 19.8 You have four samples of ideal gas, each of which contains the same number of moles of gas and has the same initial temperature, volume, and pressure. You compress each sample to one-half of its initial volume. Rank the four samples in order from highest to lowest value of the final pressure. (i) A monatomic gas compressed isothermally; (ii) a monatomic gas compressed adiabatically; (iii) a diatomic gas compressed isothermally; (iv) a diatomic gas compressed adiabatically.

ANSWER

(i) and (iii) are compressed isothermally, so $pV = \text{constant}$. The volume of each sample decreases to one-half of its initial value, so the final pressure is twice the initial pressure. Samples (ii) and (iv) are compressed adiabatically, so $pV^\gamma = \text{constant}$ and the pressure increases by a factor of 2^γ . Sample (ii) is a monatomic gas for which $\gamma = \frac{5}{3}$, so its final pressure is $2^{5/3} = 3.17$ times greater than the initial pressure. Sample (iv) is a diatomic gas for which $\gamma = \frac{7}{5}$, so its final pressure is greater than the initial pressure by a factor of $2^{7/5} = 2.64$.

CHAPTER 19 SUMMARY

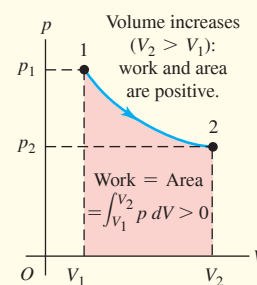
Heat and work in thermodynamic processes: A thermodynamic system has the potential to exchange energy with its surroundings by heat transfer or by mechanical work. When a system at pressure p changes volume from V_1 to V_2 , it does an amount of work W given by the integral of p with respect to volume. If the pressure is constant, the work done is equal to p times the change in volume. A negative value of W means that work is done on the system. (See Example 19.1.)

In any thermodynamic process, the heat added to the system and the work done by the system depend not only on the initial and final states, but also on the path (the series of intermediate states through which the system passes).

$$W = \int_{V_1}^{V_2} p \, dV \quad (19.2)$$

$$W = p(V_2 - V_1) \quad (19.3)$$

(constant pressure only)



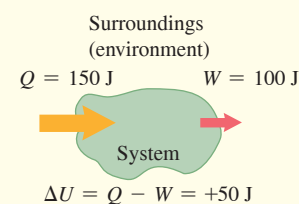
The first law of thermodynamics: The first law of thermodynamics states that when heat Q is added to a system while the system does work W , the internal energy U changes by an amount equal to $Q - W$. This law can also be expressed for an infinitesimal process. (See Examples 19.2, 19.3, and 19.5.)

The internal energy of any thermodynamic system depends only on its state. The change in internal energy in any process depends only on the initial and final states, not on the path. The internal energy of an isolated system is constant. (See Example 19.4.)

$$\Delta U = Q - W \quad (19.4)$$

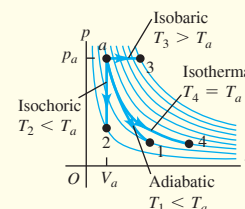
$$dU = dQ - dW \quad (19.6)$$

(infinitesimal process)



Important kinds of thermodynamic processes:

- Adiabatic process: No heat transfer into or out of a system; $Q = 0$.
- Isochoric process: Constant volume; $W = 0$.
- Isobaric process: Constant pressure; $W = p(V_2 - V_1)$.
- Isothermal process: Constant temperature.

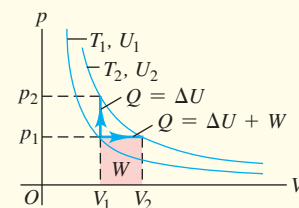


Thermodynamics of ideal gases: The internal energy of an ideal gas depends only on its temperature, not on its pressure or volume. For other substances the internal energy generally depends on both pressure and temperature.

The molar heat capacities C_V and C_p of an ideal gas differ by R , the ideal-gas constant. The dimensionless ratio of heat capacities, C_p/C_V , is denoted by γ . (See Example 19.6.)

$$C_p = C_V + R \quad (19.17)$$

$$\gamma = \frac{C_p}{C_V} \quad (19.18)$$

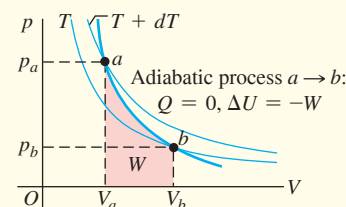


Adiabatic processes in ideal gases: For an adiabatic process for an ideal gas, the quantities $TV^{\gamma-1}$ and pV^γ are constant. The work done by an ideal gas during an adiabatic expansion can be expressed in terms of the initial and final values of temperature, or in terms of the initial and final values of pressure and volume. (See Example 19.7.)

$$W = nC_V(T_1 - T_2) \quad (19.25)$$

$$= \frac{C_V}{R}(p_1 V_1 - p_2 V_2) \quad (19.26)$$

$$= \frac{1}{\gamma - 1}(p_1 V_1 - p_2 V_2)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 19.2, 19.3, 19.4, and 19.5** (Section 19.4) before attempting these problems.

VP19.5.1 Consider the following thermodynamic processes. (a) The internal energy of a quantity of gas increases by 2.50×10^3 J as the gas does 3.25×10^3 J of work on its surroundings. How much heat flows into the gas? (b) Ice enclosed in a container does 2.50×10^4 J of work on its container as 7.00×10^3 J of heat flows out of the ice. What is the internal energy change of the ice? (c) The internal energy of a metal block increases by 4.25×10^3 J as 2.40×10^3 J of heat flows into the block. How much work did the block do on its surroundings?

VP19.5.2 How much work does a quantity of gas (volume V , pressure p) do in each of these processes? (a) V increases from 2.00×10^{-3} m³ to 4.50×10^{-3} m³ while p is a constant 6.20×10^4 Pa. (b) V is a constant 2.00×10^{-3} m³ while p increases from 6.00×10^4 Pa to 9.00×10^4 Pa. (c) V decreases from 6.00×10^{-3} m³ to 3.00×10^{-3} m³ while p is a constant 1.25×10^5 Pa. (d) V increases from 2.00×10^{-3} m³ to 4.50×10^{-3} m³ while p increases from 1.50×10^5 Pa to 5.50×10^5 Pa. The graph of this process on a pV -diagram is a straight line.

VP19.5.3 A quantity of gas is enclosed in a cylinder that has a movable piston. The gas has volume $V = 3.60 \times 10^{-3}$ m³ and pressure $p = 2.00 \times 10^5$ Pa. (a) First V decreases to 2.40×10^{-3} m³ while p remains constant; then V remains constant while p increases to 6.00×10^5 Pa. During this process 1.56×10^3 J of heat flows into the gas. For this process, how much work does the gas do, and what is its internal energy change? (b) The gas returns to the same initial volume and pressure. Now p increases to 6.00×10^5 Pa while V remains constant; then V decreases to 2.40×10^{-3} m³ while p remains constant. For this process, how much work does the gas do, and how much heat flows into the gas?

VP19.5.4 At an atmospheric pressure of 1.01×10^5 Pa, carbon dioxide (CO₂) sublimates at a temperature of -78.5°C . The heat of sublimation is 5.71×10^5 J/kg. You let heat flow into 6.0 kg of solid CO₂ (volume 4.0×10^{-3} m³) at atmospheric pressure and a constant -78.5°C until it has completely sublimed to gas at -78.5°C . The CO₂ gas occupies a volume of 3.4 m³. For this process, calculate (a) how much heat flows into the CO₂, (b) how much work the CO₂ does, and (c) the internal energy change of the CO₂.

Be sure to review **EXAMPLE 19.6** (Section 19.7) before attempting these problems.

VP19.6.1 For neon gas, $C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$. Calculate the change in internal energy of 4.00 mol of neon if its temperature (a) increases from 20.0°C to 40.0°C at constant volume; (b) increases from 15.0°C to

30.0°C at constant pressure; (c) increases from 12.0°C to 24.0°C while both the pressure and volume increase.

VP19.6.2 For nitrogen gas, $C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$. You have 2.10 mol of nitrogen gas at temperature 27.0°C and pressure $p = 1.00 \times 10^5$ Pa. (a) Calculate the initial volume V of the gas. (b) Calculate the final temperature (in $^\circ\text{C}$) of the gas and its internal energy change if (i) p doubles while V remains constant, (ii) V doubles while p remains constant, and (iii) both p and V double.

VP19.6.3 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure $p = 1.20 \times 10^5$ Pa and occupies volume $V = 0.250$ m³. Calculate the change in internal energy of the gas if the final pressure and volume of the gas are (a) $p = 2.40 \times 10^5$ Pa, $V = 0.250$ m³; (b) $p = 1.20 \times 10^5$ Pa, $V = 0.125$ m³; (c) $p = 1.80 \times 10^5$ Pa, $V = 0.600$ m³.

VP19.6.4 A quantity of n moles of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature T . You increase the absolute temperature to $2T$. Find the change in internal energy of the gas, the heat flow into the gas, and the work done by the gas if the process you used to increase the temperature is (a) isochoric, (b) isobaric, or (c) adiabatic.

Be sure to review **EXAMPLE 19.7** (Section 19.8) before attempting these problems.

VP19.7.1 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure 4.00×10^5 Pa and occupies volume 2.00×10^{-3} m³. Then the gas expands adiabatically to a new volume 6.00×10^{-3} m³. Calculate (a) the value of γ for argon, (b) the final pressure of the gas, and (c) the work done by the gas as it expands.

VP19.7.2 A quantity of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature 325 K, is under pressure 1.00×10^5 Pa, and occupies volume 6.50×10^{-3} m³. After you compress this gas adiabatically, its absolute temperature is 855 K. Calculate (a) the value of γ for oxygen, (b) the final volume of the gas, (c) the final pressure of the gas, and (d) the work done by the gas as it is compressed.

VP19.7.3 Initially 5.00 mol of neon gas ($C_V = \frac{3}{2}R$ and $\gamma = \frac{5}{2}$) are at absolute temperature 305 K and occupy volume 4.00×10^{-2} m³. Then the gas expands adiabatically to a new volume of 9.00×10^{-2} m³. Calculate (a) the initial pressure of the gas, (b) the final pressure of the gas, (c) the final temperature of the gas, and (d) the work done by the gas as it expands.

VP19.7.4 You have 1.25 mol of hydrogen gas ($C_V = \frac{5}{2}R$ and $\gamma = \frac{7}{2}$) at absolute temperature 325 K. You allow the gas to expand adiabatically to a final temperature of 195 K. (a) How much work does the gas do while being compressed? (b) What is the ratio of its final volume to its initial volume? (c) What is the ratio of the final gas pressure to the initial gas pressure?

BRIDGING PROBLEM Work Done by a van der Waals Gas

The van der Waals equation of state, an approximate representation of the behavior of gases at high pressure, is given by Eq. (18.7): $[p + (an^2/V^2)](V - nb) = nRT$, where a and b are constants having different values for different gases. (In the special case of $a = b = 0$, this is the ideal-gas equation.) (a) Calculate the work done by a gas with this equation of state in an isothermal expansion from V_1 to V_2 . (b) For ethane gas (C₂H₆), $a = 0.554$ J·m³/mol² and $b = 6.38 \times 10^{-5}$ m³/mol. Calculate the work W done by 1.80 mol of ethane when it expands from 2.00×10^{-3} m³ to 4.00×10^{-3} m³ at

a constant temperature of 300 K. Do the calculation by using (i) the van der Waals equation of state and (ii) the ideal-gas equation of state. (c) For which equation of state is W larger? Why should this be so?

SOLUTION GUIDE

IDENTIFY and SET UP

- Review the discussion of the van der Waals equation of state in Section 18.1. What is the significance of the quantities a and b ?

Continued

- Decide how to find the work done by an expanding gas whose pressure p does not depend on V in the same way as for an ideal gas. (Hint: See Section 19.2.)
- How will you find the work done by an expanding ideal gas?

EXECUTE

- Find the general expression for the work done by a van der Waals gas as it expands from volume V_1 to volume V_2 (Fig. 19.22). (Hint: If you set $a = b = 0$ in your result, it should reduce to the expression for the work done by an expanding ideal gas.)
- Use your result from step 4 to solve part (b) for ethane treated as a van der Waals gas.
- Use the formula you chose in step 3 to solve part (b) for ethane treated as an ideal gas.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q19.1 For the following processes, is the work done by the system (defined as the expanding or contracting gas) on the environment positive or negative? (a) expansion of the burned gasoline–air mixture in the cylinder of an automobile engine; (b) opening a bottle of champagne; (c) filling a scuba tank with compressed air; (d) partial crumpling of a sealed, empty water bottle as you drive from the mountains down to sea level.

Q19.2 It is not correct to say that an object contains a certain amount of heat, yet an object can transfer heat to another object. How can an object give away something it does not have in the first place?

Q19.3 In which situation must you do more work: inflating a balloon at sea level or inflating the same balloon to the same volume at the summit of Mt. McKinley? Explain in terms of pressure and volume change.

Q19.4 If you are told the initial and final states of a system and the associated change in internal energy, can you determine whether the internal energy change was due to work or to heat transfer? Explain.

Q19.5 Discuss the application of the first law of thermodynamics to a mountaineer who eats food, gets warm and perspires a lot during a climb, and does a lot of mechanical work in raising herself to the summit. The mountaineer also gets warm during the descent. Is the source of this energy the same as the source during the ascent?

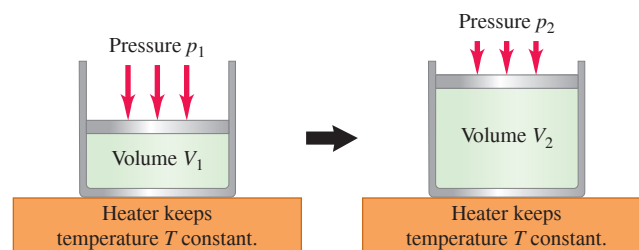
Q19.6 When ice melts at 0°C , its volume decreases. Is the internal energy change greater than, less than, or equal to the heat added? How can you tell?

Q19.7 You hold an inflated balloon over a hot-air vent in your house and watch it slowly expand. You then remove it and let it cool back to room temperature. During the expansion, which was larger: the heat added to the balloon or the work done by the air inside it? Explain. (Assume that air is an ideal gas.) Once the balloon has returned to room temperature, how does the net heat gained or lost by the air inside it compare to the net work done on or by the surrounding air?

Q19.8 You bake chocolate chip cookies and put them, still warm, in a container with a loose (not airtight) lid. What kind of process does the air inside the container undergo as the cookies gradually cool to room temperature (isothermal, isochoric, adiabatic, isobaric, or some combination)? Explain.

Q19.9 Imagine a gas made up entirely of negatively charged electrons. Like charges repel, so the electrons exert repulsive forces on each other. Would you expect that the temperature of such a gas would rise, fall, or stay the same in a free expansion? Why?

Figure 19.22 A gas undergoes an isothermal expansion.

**EVALUATE**

- Is the difference between W for the two equations of state large enough to be significant?
- Does the term with a in the van der Waals equation of state increase or decrease the amount of work done? What about the term with b ? Which one is more important for the ethane in this problem?

Q19.10 In an adiabatic process for an ideal gas, the pressure decreases. In this process does the internal energy of the gas increase or decrease? Explain.

Q19.11 When you blow on the back of your hand with your mouth wide open, your breath feels warm. But if you partially close your mouth to form an “o” and then blow on your hand, your breath feels cool. Why?

Q19.12 An ideal gas expands while the pressure is kept constant. During this process, does heat flow into the gas or out of the gas? Justify your answer.

Q19.13 A liquid is irregularly stirred in a well-insulated container and thereby undergoes a rise in temperature. Regard the liquid as the system. Has heat been transferred? How can you tell? Has work been done? How can you tell? Why is it important that the stirring is irregular? What is the sign of ΔU ? How can you tell?

Q19.14 When you use a hand pump to inflate the tires of your bicycle, the pump gets warm after a while. Why? What happens to the temperature of the air in the pump as you compress it? Why does this happen? When you raise the pump handle to draw outside air into the pump, what happens to the temperature of the air taken in? Again, why does this happen?

Q19.15 In the carburetor of an aircraft or automobile engine, air flows through a relatively small aperture and then expands. In cool, foggy weather, ice sometimes forms in this aperture even though the outside air temperature is above freezing. Why?

Q19.16 On a sunny day, large “bubbles” of air form on the sunwarmed earth, gradually expand, and finally break free to rise through the atmosphere. Soaring birds and glider pilots are fond of using these “thermals” to gain altitude easily. This expansion is essentially an adiabatic process. Why?

Q19.17 The prevailing winds on the Hawaiian island of Kauai blow from the northeast. The winds cool as they go up the slope of Mt. Waialeale (elevation 1523 m), causing water vapor to condense and rain to fall. There is much more precipitation at the summit than at the base of the mountain. In fact, Mt. Waialeale is the rainiest spot on earth, averaging 11.7 m of rainfall a year. But what makes the winds cool?

Q19.18 Applying the same considerations as in Question Q19.17, explain why the island of Niihau, a few kilometers to the southwest of Kauai, is almost a desert and farms there need to be irrigated.

Q19.19 In a constant-volume process, $dU = nC_V dT$. But in a constant-pressure process, it is *not* true that $dU = nC_p dT$. Why not?

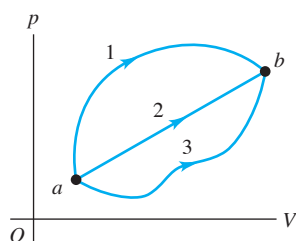
Q19.20 When a gas surrounded by air is compressed adiabatically, its temperature rises even though there is no heat input to the gas. Where does the energy come from to raise the temperature?

Q19.21 When a gas expands adiabatically, it does work on its surroundings. But if there is no heat input to the gas, where does the energy come from to do the work?

Q19.22 The gas used in separating the two uranium isotopes ^{235}U and ^{238}U has the formula UF_6 . If you added heat at equal rates to a mole of UF_6 gas and a mole of H_2 gas, which one's temperature would you expect to rise faster? Explain.

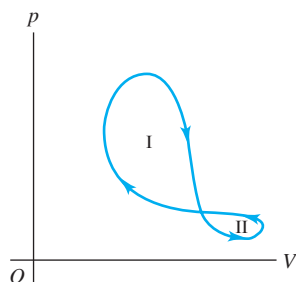
Q19.23 A system is taken from state a to state b along the three paths shown in **Fig. Q19.23**. (a) Along which path is the work done by the system the greatest? The least? (b) If $U_b > U_a$, along which path is the absolute value of the heat transfer, $|Q|$, the greatest? For this path, is heat absorbed or liberated by the system? Explain.

Figure Q19.23



Q19.24 A thermodynamic system undergoes a cyclic process as shown in **Fig. Q19.24**. The cycle consists of two closed loops: I and II. (a) Over one complete cycle, does the system do positive or negative work? (b) In each loop, is the net work done by the system positive or negative? (c) Over one complete cycle, does heat flow into or out of the system? (d) In each loop, does heat flow into or out of the system? Explain.

Figure Q19.24



EXERCISES

Section 19.2 Work Done During Volume Changes and

Section 19.3 Paths Between Thermodynamic States

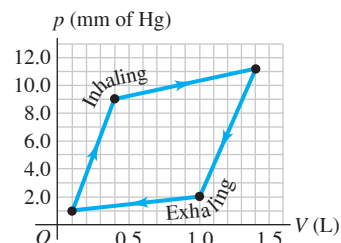
19.1 • Two moles of an ideal gas are heated at constant pressure from $T = 27^\circ\text{C}$ to $T = 107^\circ\text{C}$. (a) Draw a pV -diagram for this process. (b) Calculate the work done by the gas.

19.2 • Six moles of an ideal gas are in a cylinder fitted at one end with a movable piston. The initial temperature of the gas is 27.0°C and the pressure is constant. As part of a machine design project, calculate the final temperature of the gas after it has done $2.40 \times 10^3 \text{ J}$ of work.

19.3 • **CALC** Two moles of an ideal gas are compressed in a cylinder at a constant temperature of 65.0°C until the original pressure has tripled. (a) Sketch a pV -diagram for this process. (b) Calculate the amount of work done.

19.4 • **BIO Work Done by the Lungs.** The graph in **Fig. E19.4** shows a pV -diagram of the air in a human lung when a person is inhaling and then exhaling a deep breath. Such graphs, obtained in clinical practice, are normally somewhat curved, but we have modeled one as a set of straight lines of the same general shape. (*Important:* The pressure shown is the *gauge* pressure, *not* the absolute pressure.) (a) How many joules of *net* work does this person's lung do during one complete breath? (b) The process illustrated here is somewhat different from those we have been studying, because the pressure change is due to changes in the amount of gas in the lung, not to temperature changes. (Think of your own breathing. Your lungs do not expand because they've gotten hot.) If the temperature of the air in the lung remains a reasonable 20°C , what is the maximum number of moles in this person's lung during a breath?

Figure E19.4



19.5 • **CALC** During the time 0.305 mol of an ideal gas undergoes an isothermal compression at 22.0°C , 392 J of work is done on it by the surroundings. (a) If the final pressure is 1.76 atm, what was the initial pressure? (b) Sketch a pV -diagram for the process.

19.6 • A gas undergoes two processes. In the first, the volume remains constant at 0.200 m^3 and the pressure increases from $2.00 \times 10^5 \text{ Pa}$ to $5.00 \times 10^5 \text{ Pa}$. The second process is a compression to a volume of 0.120 m^3 at a constant pressure of $5.00 \times 10^5 \text{ Pa}$. (a) In a pV -diagram, show both processes. (b) Find the total work done by the gas during both processes.

19.7 • **Work Done in a Cyclic Process.** (a) In **Fig. 19.7a**, consider the closed loop $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$. This is a *cyclic* process in which the initial and final states are the same. Find the total work done by the system in this cyclic process, and show that it is equal to the area enclosed by the loop. (b) How is the work done for the process in part (a) related to the work done if the loop is traversed in the opposite direction, $1 \rightarrow 4 \rightarrow 2 \rightarrow 3 \rightarrow 1$? Explain.

Section 19.4 Internal Energy and the First Law of Thermodynamics

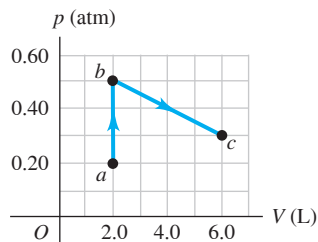
19.8 • **CALC** An ideal gas undergoes a process during which the pressure is kept directly proportional to the volume, so that $p = \alpha V$, where α is a positive constant. If the volume changes from V_1 to V_2 , how much work is done by the gas? Express your answer in terms of V_1 , V_2 , and α .

19.9 • A gas in a cylinder expands from a volume of 0.110 m^3 to 0.320 m^3 . Heat flows into the gas just rapidly enough to keep the pressure constant at $1.65 \times 10^5 \text{ Pa}$ during the expansion. The total heat added is $1.15 \times 10^5 \text{ J}$. (a) Find the work done by the gas. (b) Find the change in internal energy of the gas. (c) Does it matter whether the gas is ideal? Why or why not?

19.10 • Five moles of an ideal monatomic gas with an initial temperature of 127°C expand and, in the process, absorb 1500 J of heat and do 2100 J of work. What is the final temperature of the gas?

19.11 •• The process abc shown in the pV -diagram in **Fig. E19.11** involves 0.0175 mol of an ideal gas. (a) What was the lowest temperature the gas reached in this process? Where did it occur? (b) How much work was done by or on the gas from a to b ? From b to c ? (c) If 215 J of heat was put into the gas during abc , how many of those joules went into internal energy?

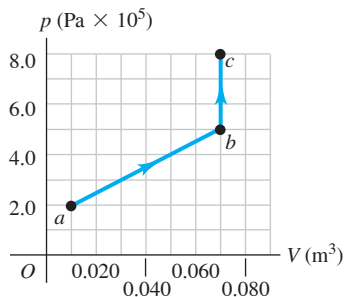
Figure E19.11



19.12 • A gas in a cylinder is held at a constant pressure of 1.80×10^5 Pa and is cooled and compressed from 1.70 m^3 to 1.20 m^3 . The internal energy of the gas decreases by 1.40×10^5 J. (a) Find the work done by the gas. (b) Find the absolute value of the heat flow, $|Q|$, into or out of the gas, and state the direction of the heat flow. (c) Does it matter whether the gas is ideal? Why or why not?

19.13 •• The pV -diagram in **Fig. E19.13** shows a process abc involving 0.450 mol of an ideal gas. (a) What was the temperature of this gas at points a , b , and c ? (b) How much work was done by or on the gas in this process? (c) How much heat had to be added during the process to increase the internal energy of the gas by 15,000 J?

Figure E19.13



19.14 • Boiling Water at High Pressure. When water is boiled at a pressure of 2.00 atm, the heat of vaporization is 2.20×10^6 J/kg and the boiling point is 120°C . At this pressure, 1.00 kg of water has a volume of $1.00 \times 10^{-3} \text{ m}^3$, and 1.00 kg of steam has a volume of 0.824 m^3 . (a) Compute the work done when 1.00 kg of steam is formed at this temperature. (b) Compute the increase in internal energy of the water.

19.15 • A thermodynamic system undergoes processes in which $|Q| = 100 \text{ J}$ and $|W| = 300 \text{ J}$. Find Q and W , including whether the quantity is positive or negative, if the change in internal energy is (a) +400 J, (b) +200 J, (c) -200 J, and (d) -400 J.

Section 19.5 Kinds of Thermodynamic Processes,

Section 19.6 Internal Energy of an Ideal Gas, and

Section 19.7 Heat Capacities of an Ideal Gas

19.16 • During an isothermal compression of an ideal gas, 410 J of heat must be removed from the gas to maintain constant temperature. How much work is done by the gas during the process?

19.17 • A cylinder contains 0.250 mol of carbon dioxide (CO_2) gas at a temperature of 27.0°C . The cylinder is provided with a frictionless piston, which maintains a constant pressure of 1.00 atm on the gas. The gas is heated until its temperature increases to 127.0°C . Assume that the CO_2 may be treated as an ideal gas. (a) Draw a pV -diagram for this process. (b) How much work is done by the gas in this process? (c) On what is this work done? (d) What is the change in internal energy of the gas? (e) How much heat was supplied to the gas? (f) How much work would have been done if the pressure had been 0.50 atm?

19.18 • A cylinder contains 0.0100 mol of helium at $T = 27.0^\circ\text{C}$. (a) How much heat is needed to raise the temperature to 67.0°C while keeping the volume constant? Draw a pV -diagram for this process. (b) If instead the pressure of the helium is kept constant, how much heat is needed to raise the temperature from 27.0°C to 67.0°C ? Draw a pV -diagram for this process. (c) What accounts for the difference between your answers to parts (a) and (b)? In which case is more heat required? What becomes of the additional heat? (d) If the gas is ideal, what is the change in its internal energy in part (a)? In part (b)? How do the two answers compare? Why?

19.19 • In an experiment to simulate conditions inside an automobile engine, 0.185 mol of air at 780 K and 3.00×10^6 Pa is contained in a cylinder of volume 40.0 cm^3 . Then 645 J of heat is transferred to the cylinder. (a) If the volume of the cylinder is constant while the heat is added, what is the final temperature of the air? Assume that the air is essentially nitrogen gas, and use the data in Table 19.1 even though the pressure is not low. Draw a pV -diagram for this process. (b) If instead the volume of the cylinder is allowed to increase while the pressure remains constant, repeat part (a).

19.20 •• A quantity of 5.00 mol of a monatomic ideal gas has an initial pressure of $p_a = 1.00 \times 10^5$ Pa and an initial volume of $V_a = 0.0200 \text{ m}^3$. Two paths are used to take the gas to a final state for which $p_b = 1.80 \times 10^5$ Pa and $V_b = 0.0280 \text{ m}^3$. (a) Along path 1 the pressure is increased to $p_a = 1.80 \times 10^5$ Pa while the volume is kept constant with a value of 0.0200 m^3 . Then the volume is increased to 0.0280 m^3 while the pressure is kept at a constant 1.80×10^5 Pa. Calculate the total Q and total W for this path. (b) Along path 2 the volume is increased to 0.0280 m^3 while the pressure remains 1.00×10^5 Pa and then the pressure is increased to 1.80×10^5 Pa while the volume remains constant at 0.0280 m^3 . Calculate the total Q and total W for this path. (c) Calculate the quantity $Q - W$ for each path. Your results should show that for these processes Q and W are path dependent but $Q - W$ is path independent.

19.21 • Heat Q flows into a monatomic ideal gas, and the volume increases while the pressure is kept constant. What fraction of the heat energy is used to do the expansion work of the gas?

19.22 • Three moles of an ideal monatomic gas expands at a constant pressure of 2.50 atm; the volume of the gas changes from $3.20 \times 10^{-2} \text{ m}^3$ to $4.50 \times 10^{-2} \text{ m}^3$. Calculate (a) the initial and final temperatures of the gas; (b) the amount of work the gas does in expanding; (c) the amount of heat added to the gas; (d) the change in internal energy of the gas.

19.23 • An experimenter adds 970 J of heat to 1.75 mol of an ideal gas to heat it from 10.0°C to 25.0°C at constant pressure. The gas does +223 J of work during the expansion. (a) Calculate the change in internal energy of the gas. (b) Calculate γ for the gas.

19.24 • A quantity of 2.00 mol of a monatomic ideal gas undergoes a compression during which the volume decreases from 0.0800 m^3 to 0.0500 m^3 while the pressure stays constant at a value of 1.80×10^4 Pa. (a) What is the work W ? Is work done by the gas or on the gas? (b) What is the heat flow Q ? Does heat enter or leave the gas? (c) What is the internal energy change for the gas? Does the internal energy of the gas increase or decrease?

19.25 • CALC The temperature of 0.150 mol of an ideal gas is held constant at 77.0°C while its volume is reduced to 25.0% of its initial volume. The initial pressure of the gas is 1.25 atm. (a) Determine the work done by the gas. (b) What is the change in its internal energy? (c) Does the gas exchange heat with its surroundings? If so, how much? Does the gas absorb or liberate heat?

19.26 •• For an ideal gas C_V and C_p are different because of the work W associated with the volume change for a constant-pressure process. To explore the difference between C_V and C_p for a liquid or a solid, consider the process in which 5.00 mol of ethanol is warmed from 10.0°C to 60.0°C while the applied pressure remains a constant 1.00 atm. The molar mass of ethanol is $M = 46.1\text{ g/mol}$. (a) What is the mass of 5.00 mol of ethanol? (b) How much heat Q enters the ethanol? (c) How much work W is done by the ethanol because of its thermal expansion? (d) What is the ratio W/Q ?

Section 19.8 Adiabatic Processes for an Ideal Gas

19.27 • A monatomic ideal gas that is initially at $1.50 \times 10^5\text{ Pa}$ and has a volume of 0.0800 m^3 is compressed adiabatically to a volume of 0.0400 m^3 . (a) What is the final pressure? (b) How much work is done by the gas? (c) What is the ratio of the final temperature of the gas to its initial temperature? Is the gas heated or cooled by this compression?

19.28 •• Five moles of monatomic ideal gas have initial pressure $2.50 \times 10^3\text{ Pa}$ and initial volume 2.10 m^3 . While undergoing an adiabatic expansion, the gas does 1480 J of work. What is the final pressure of the gas after the expansion?

19.29 • During an adiabatic expansion the temperature of 0.450 mol of argon (Ar) drops from 66.0°C to 10.0°C . The argon may be treated as an ideal gas. (a) Draw a pV -diagram for this process. (b) How much work does the gas do? (c) What is the change in internal energy of the gas?

19.30 •• A player bounces a basketball on the floor, compressing it to 80.0% of its original volume. The air (assume it is essentially N_2 gas) inside the ball is originally at 20.0°C and 2.00 atm. The ball's inside diameter is 23.9 cm. (a) What temperature does the air in the ball reach at its maximum compression? Assume the compression is adiabatic and treat the gas as ideal. (b) By how much does the internal energy of the air change between the ball's original state and its maximum compression?

19.31 •• On a warm summer day, a large mass of air (atmospheric pressure $1.01 \times 10^5\text{ Pa}$) is heated by the ground to 26.0°C and then begins to rise through the cooler surrounding air. (This can be treated approximately as an adiabatic process; why?) Calculate the temperature of the air mass when it has risen to a level at which atmospheric pressure is only $0.850 \times 10^5\text{ Pa}$. Assume that air is an ideal gas, with $\gamma = 1.40$. (This rate of cooling for dry, rising air, corresponding to roughly 1°C per 100 m of altitude, is called the *dry adiabatic lapse rate*.)

19.32 • A cylinder contains 0.100 mol of an ideal monatomic gas. Initially the gas is at $1.00 \times 10^5\text{ Pa}$ and occupies a volume of $2.50 \times 10^{-3}\text{ m}^3$. (a) Find the initial temperature of the gas in kelvins. (b) If the gas is allowed to expand to twice the initial volume, find the final temperature (in kelvins) and pressure of the gas if the expansion is (i) isothermal; (ii) isobaric; (iii) adiabatic.

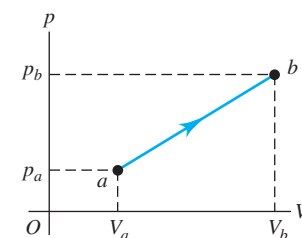
19.33 •• Helium gas undergoes an adiabatic process in which the Kelvin temperature doubles. By what factor does the pressure change?

19.34 • For a diatomic ideal gas, 3.01×10^{20} molecules undergo an adiabatic process in which the temperature increases from 25.0°C to 60.0°C . What is the change in the internal energy of the gas?

PROBLEMS

19.35 •• A quantity of air is taken from state a to state b along a path that is a straight line in the pV -diagram (Fig. P19.35). (a) In this process, does the temperature of the gas increase, decrease, or stay the same? Explain. (b) If $V_a = 0.0700\text{ m}^3$, $V_b = 0.1100\text{ m}^3$, $p_a = 1.00 \times 10^5\text{ Pa}$, and $p_b = 1.40 \times 10^5\text{ Pa}$, what is the work W done by the gas in this process? Assume that the gas may be treated as ideal.

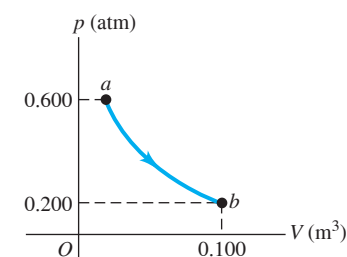
Figure P19.35



19.36 •• When water boils in a pot with a loose lid, the increased pressure forces the lid upward, releasing steam and reducing the pressure in the pot so that the lid drops back down. If the water continues to boil, the lid can rattle upward and downward. Consider a 1 L pot half filled with boiling water. In this case there is half a liter, or $\frac{1}{11.2}$ mol, of air and steam above the water. (a) Estimate the mass of the lid. (b) When the water is rapidly boiling, estimate the height to which the lid jumps during a steam-release event. (c) By multiplying the weight of the lid by the height of the jump, estimate the work done by the steam as the lid moves upward. (d) Treat the process as adiabatic and use the first law of thermodynamics to estimate the drop in the temperature of the steam as it pushes the lid upward.

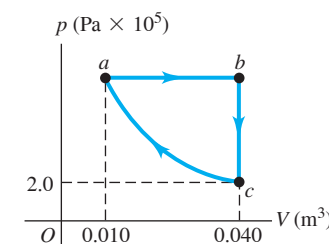
19.37 •• Figure P19.37 shows the pV -diagram for a process in which the temperature of the ideal gas remains constant at 85°C . (a) How many moles of gas are involved? (b) What volume does this gas occupy at a ? (c) How much work was done by or on the gas from a to b ? (d) By how much did the internal energy of the gas change during this process?

Figure P19.37



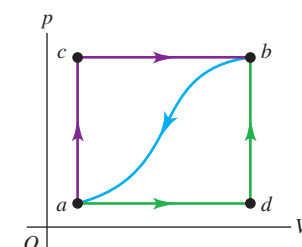
19.38 •• The graph in Fig. P19.38 shows a pV -diagram for 3.25 mol of ideal helium (He) gas. Part ca of this process is isothermal. (a) Find the pressure of the He at point a . (b) Find the temperature of the He at points a , b , and c . (c) How much heat entered or left the He during segments ab , bc , and ca ? In each segment, did the heat enter or leave? (d) By how much did the internal energy of the He change from a to b , from b to c , and from c to a ? Indicate whether this energy increased or decreased.

Figure P19.38



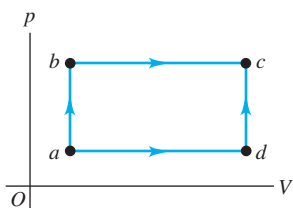
19.39 •• When a system is taken from state a to state b in Fig. P19.39 along path acb , 90.0 J of heat flows into the system and 60.0 J of work is done by the system. (a) How much heat flows into the system along path adb if the work done by the system is 15.0 J? (b) When the system is returned from b to a along the curved path, the absolute value of the work done by the system is 35.0 J. Does the system absorb or liberate heat? How much heat? (c) If $U_a = 0$ and $U_d = 8.0\text{ J}$, find the heat absorbed in the processes ad and db .

Figure P19.39



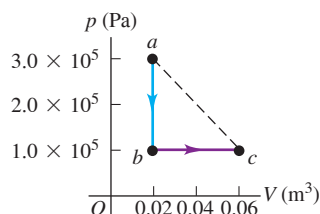
19.40 • A thermodynamic system is taken from state a to state c in **Fig. P19.40** along either path abc or path adc . Along path abc , the work W done by the system is 450 J. Along path adc , W is 120 J. The internal energies of each of the four states shown in the figure are $U_a = 150$ J, $U_b = 240$ J, $U_c = 680$ J, and $U_d = 330$ J. Calculate the heat flow Q for each of the four processes ab , bc , ad , and dc . In each process, does the system absorb or liberate heat?

Figure P19.40



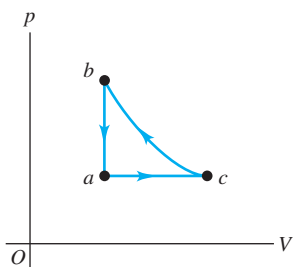
19.41 • A volume of air (assumed to be an ideal gas) is first cooled without changing its volume and then expanded without changing its pressure, as shown by path abc in **Fig. P19.41**. (a) How does the final temperature of the gas compare with its initial temperature? (b) How much heat does the air exchange with its surroundings during process abc ? Does the air absorb heat or release heat during this process? Explain. (c) If the air instead expands from state a to state c by the straight-line path shown, how much heat does it exchange with its surroundings?

Figure P19.41



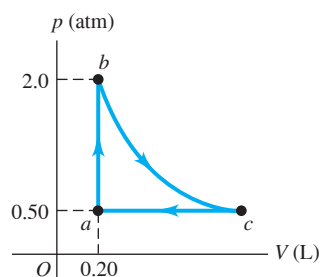
19.42 •• Three moles of an ideal gas are taken around cycle acb shown in **Fig. P19.42**. For this gas, $C_p = 29.1$ J/mol·K. Process ac is at constant pressure, process ba is at constant volume, and process cb is adiabatic. The temperatures of the gas in states a , c , and b are $T_a = 300$ K, $T_c = 492$ K, and $T_b = 600$ K. Calculate the total work W for the cycle.

Figure P19.42



19.43 •• Figure P19.43 shows a pV -diagram for 0.0040 mol of ideal H_2 gas. The temperature of the gas does not change during segment bc . (a) What volume does this gas occupy at point c ? (b) Find the temperature of the gas at points a , b , and c . (c) How much heat went into or out of the gas during segments ab , ca , and bc ? Indicate whether the heat has gone into or out of the gas.

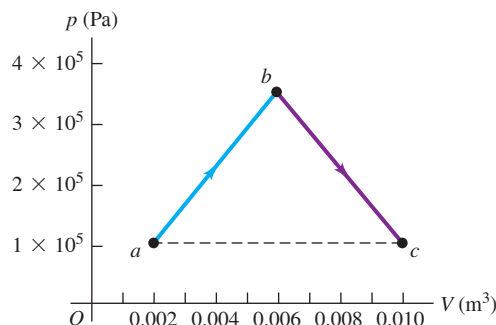
Figure P19.43



(d) Find the change in the internal energy of this hydrogen during segments ab , bc , and ca . Indicate whether the internal energy increased or decreased during each segment.

19.44 •• (a) One-third of a mole of He gas is taken along the path abc shown in **Fig. P19.44**. Assume that the gas may be treated as ideal. How much heat is transferred into or out of the gas? (b) If the gas instead went directly from state a to state c along the horizontal dashed line in **Fig. P19.44**, how much heat would be transferred into or out of the gas? (c) How does Q in part (b) compare with Q in part (a)? Explain.

Figure P19.44



19.45 •• DATA You place a quantity of gas into a metal cylinder that has a movable piston at one end. No gas leaks out of the cylinder as the piston moves. The external force applied to the piston can be varied to change the gas pressure as you move the piston to change the volume of the gas. A pressure gauge attached to the interior wall of the cylinder measures the gas pressure, and you can calculate the volume of the gas from a measurement of the piston's position in the cylinder.

You start with a pressure of 1.0 atm and a gas volume of 3.0 L. Holding the pressure constant, you increase the volume to 5.0 L. Then, keeping the volume constant at 5.0 L, you increase the pressure to 3.0 atm. Next you decrease the pressure linearly as a function of volume until the volume is 3.0 L and the pressure is 2.0 atm. Finally, you keep the volume constant at 3.0 L and decrease the pressure to 1.0 atm, returning the gas to its initial pressure and volume. The walls of the cylinder are good conductors of heat, and you provide the required heat sources and heat sinks so that the necessary heat flows can occur. At these relatively high pressures, you suspect that the ideal-gas equation will not apply with much accuracy. You don't know what gas is in the cylinder or whether it is monatomic, diatomic, or polyatomic. (a) Plot the cycle in the pV -plane. (b) What is the net heat flow for the gas during this cycle? Is there net heat flow into or out of the gas?

19.46 •• Nitrogen gas in an expandable container is cooled from 50.0°C to 10.0°C with the pressure held constant at 3.00×10^5 Pa. The total heat liberated by the gas is 2.50×10^4 J. Assume that the gas may be treated as ideal. Find (a) the number of moles of gas; (b) the change in internal energy of the gas; (c) the work done by the gas. (d) How much heat would be liberated by the gas for the same temperature change if the volume were constant?

19.47 • CALC A cylinder with a frictionless, movable piston like that shown in **Fig. 19.5** contains a quantity of helium gas. Initially the gas is at 1.00×10^5 Pa and 300 K and occupies a volume of 1.50 L. The gas then undergoes two processes. In the first, the gas is heated and the piston is allowed to move to keep the temperature at 300 K. This continues until the pressure reaches 2.50×10^4 Pa. In the second process, the gas is compressed at constant pressure until it returns to its original volume of 1.50 L. Assume that the gas may be treated as ideal. (a) In a pV -diagram, show both processes. (b) Find the volume of the gas at the end of the first process, and the pressure and temperature at the end of the second process. (c) Find the total work done by the gas during both processes. (d) What would you have to do to the gas to return it to its original pressure and temperature?

19.48 ••• CP A steel cargo drum has a height of 880 mm and a diameter of 610 mm. With its top removed it has a mass of 17.3 kg. The drum is turned upside down at the surface of the North Atlantic and is pulled downward into the ocean by a robotic submarine. On this day the surface temperature is 23.0°C and the surface air pressure is $p_0 = 101 \text{ kPa}$. The water temperature decreases linearly with depth to 3.0°C at 1000 m below the surface. As the drum moves downward in the ocean, the air inside the drum is compressed, reducing the upward buoyant force. (a) At what depth y_{neutral} is the barrel neutrally buoyant? (*Hint:* The pressure in the drum is equal to the sea pressure, which at depth y is $p_0 + \rho gy$, where $\rho = 1025 \text{ kg/m}^3$ is the density of seawater. The temperature at depth can be determined using the information above. Together with the ideal-gas law, you can derive a formula for the volume of air at depth y , and therefore a formula for the upward buoyant force as a function of depth.) (b) What is the volume of the air in the drum at depth y_{neutral} ?

19.49 ••• Chinook. During certain seasons strong winds called chinooks blow from the west across the eastern slopes of the Rockies and downhill into Denver and nearby areas. Although the mountains are cool, the wind in Denver is very hot; within a few minutes after the chinook wind arrives, the temperature can climb 20°C (“chinook” refers to a Native American people of the Pacific Northwest). Similar winds occur in the Alps (called foehns) and in southern California (called Santa Anas). (a) Explain why the temperature of the chinook wind rises as it descends the slopes. Why is it important that the wind be fast moving? (b) Suppose a strong wind is blowing toward Denver (elevation 1630 m) from Grays Peak (80 km west of Denver, at an elevation of 4350 m), where the air pressure is $5.60 \times 10^4 \text{ Pa}$ and the air temperature is -15.0°C . The temperature and pressure in Denver before the wind arrives are 2.0°C and $8.12 \times 10^4 \text{ Pa}$. By how many Celsius degrees will the temperature in Denver rise when the chinook arrives?

19.50 ••• High-Altitude Research. Figure P19.50

A large research balloon containing $2.00 \times 10^3 \text{ m}^3$ of helium gas at 1.00 atm and a temperature of 15.0°C rises rapidly from ground level to an altitude at which the atmospheric pressure is only 0.900 atm (**Fig. P19.50**). Assume the helium behaves like an ideal gas and the balloon’s ascent is too rapid to permit much heat exchange with the surrounding air. (a) Calculate the volume of the gas at the higher altitude. (b) Calculate the temperature of the gas at the higher altitude. (c) What is the change in internal energy of the helium as the balloon rises to the higher altitude?

19.51 ••• An air pump has a cylinder 0.250 m long with a movable piston. The pump is used to compress air from the atmosphere (at absolute pressure $1.01 \times 10^5 \text{ Pa}$) into a very large tank at $3.80 \times 10^5 \text{ Pa}$ gauge pressure. (For air, $C_V = 20.8 \text{ J/mol} \cdot \text{K}$.) (a) The piston begins the compression stroke at the open end of the cylinder. How far down the length of the cylinder has the piston moved when air first begins to flow from the cylinder into the tank? Assume that the compression is adiabatic. (b) If the air is taken into the pump at 27.0°C , what is the temperature of the compressed air? (c) How much work does the pump do in putting 20.0 mol of air into the tank?

19.52 •• A certain ideal gas has molar heat capacity at constant volume C_V . A sample of this gas initially occupies a volume V_0 at pressure p_0 and absolute temperature T_0 . The gas expands isobarically to a volume $2V_0$ and then expands further adiabatically to a final volume $4V_0$. (a) Draw a pV -diagram for this sequence of processes. (b) Compute the total work done by the gas for this sequence of processes. (c) Find the

final temperature of the gas. (d) Find the absolute value of the total heat flow $|Q|$ into or out of the gas for this sequence of processes, and state the direction of heat flow.

19.53 • A monatomic ideal gas expands slowly to twice its original volume, doing 450 J of work in the process. Find the heat added to the gas and the change in internal energy of the gas if the process is (a) isothermal; (b) adiabatic; (c) isobaric.

19.54 •• CALC A cylinder with a piston contains 0.250 mol of oxygen at $2.40 \times 10^5 \text{ Pa}$ and 355 K. The oxygen may be treated as an ideal gas. The gas first expands isobarically to twice its original volume. It is then compressed isothermally back to its original volume, and finally it is cooled isochorically to its original pressure. (a) Show the series of processes on a pV -diagram. Compute (b) the temperature during the isothermal compression; (c) the maximum pressure; (d) the total work done by the piston on the gas during the series of processes.

19.55 • Use the conditions and processes of Problem 19.54 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial expansion; (b) the work done, the heat added, and the internal energy change during the final cooling; (c) the internal energy change during the isothermal compression.

19.56 •• CALC A cylinder with a piston contains 0.150 mol of nitrogen at $1.80 \times 10^5 \text{ Pa}$ and 300 K. The nitrogen may be treated as an ideal gas. The gas is first compressed isobarically to half its original volume. It then expands adiabatically back to its original volume, and finally it is heated isochorically to its original pressure. (a) Show the series of processes in a pV -diagram. (b) Compute the temperatures at the beginning and end of the adiabatic expansion. (c) Compute the minimum pressure.

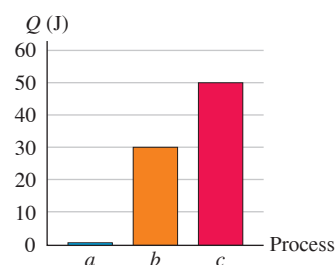
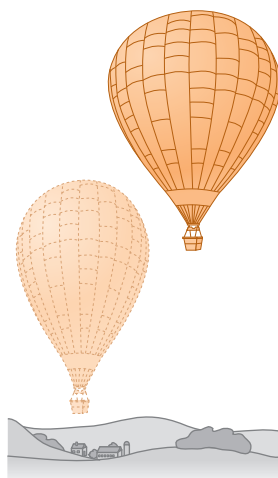
19.57 • Use the conditions and processes of Problem 19.56 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial compression; (b) the work done by the gas, the heat added to it, and its internal energy change during the adiabatic expansion; (c) the work done, the heat added, and the internal energy change during the final heating.

19.58 • Comparing Thermodynamic Processes. In a cylinder, 1.20 mol of an ideal monatomic gas, initially at $3.60 \times 10^5 \text{ Pa}$ and 300 K, expands until its volume triples. Compute the work done by the gas if the expansion is (a) isothermal; (b) adiabatic; (c) isobaric. (d) Show each process in a pV -diagram. In which case is the absolute value of the work done by the gas greatest? Least? (e) In which case is the absolute value of the heat transfer greatest? Least? (f) In which case is the absolute value of the change in internal energy of the gas greatest? Least?

19.59 •• DATA Figure P19.59

You have recorded measurements of the heat flow Q into 0.300 mol of a gas that starts at $T_1 = 20.0^{\circ}\text{C}$ and ends at a temperature T_2 . You measured Q for three processes: one isobaric, one isochoric, and one adiabatic. In each case, T_2 was the same. **Figure P19.59** summarizes your results. But you lost a page from your lab notebook and don’t have a record of the value

of T_2 ; you also don’t know which process was isobaric, isochoric, or adiabatic. Each process was done at a sufficiently low pressure for the gas to be treated as ideal. (a) Identify each process a , b , or c as isobaric, isochoric, or adiabatic. (b) What is the value of T_2 ? (c) How much work is done by the gas in each process? (d) For which process is the magnitude of the volume change the greatest? (e) For each process, does the volume of the gas increase, decrease, or stay the same?



19.60 •• DATA You compress a gas in an insulated cylinder—no heat flows into or out of the gas. The gas pressure is fairly low, so treating the gas as ideal is a good approximation. When you measure the pressure as a function of the volume of the gas, you obtain these results:

V (L)	2.50	2.02	1.48	1.01	0.50
p (atm)	0.101	0.139	0.202	0.361	0.952

(a) Graph $\log(p)$ versus $\log(V)$, with p in Pa and V in m^3 . Explain why the data points fall close to a straight line. (b) Use your graph to calculate γ for the gas. Is the gas monatomic, diatomic, or polyatomic? (c) When $p = 0.101$ atm and $V = 2.50$ L, the temperature is 22.0°C . Apply the ideal-gas equation and calculate the temperature for each of the other pairs of p and V values. In this compression, does the temperature of the gas increase, decrease, or stay constant?

19.61 ••• A pneumatic shock absorber consists of a cylinder with a radius of 1.50 cm and a length that varies from a maximum of 30.0 cm to a minimum of $30.0\text{ cm}/v$, where v is the compression ratio. For ease of manufacturing, v must be an integer. When the cylinder is fully extended and cooled, the air inside has an ambient pressure of 101 kPa and ambient temperature of 30.0°C . A quick adiabatic compression heats the air. The cylinder then expands isothermally until it reaches its maximum length. At that time the length remains fixed and the cylinder cools isochorically, returning to the ambient temperature. (a) How much work is done by the air during a full cycle, in terms of v ? (b) If the air temperature cannot exceed 400°C , what is the maximum integer value of v ? (c) If at least 25.0 J of work must be done by the gas in a full-compression cycle, what is the minimum allowable integer value of v ? (d) What is the only permissible value of v ? (e) How much heat leaves the air in the cylinder during the isochoric process if v has the value calculated in (d)?

CHALLENGE PROBLEM

19.62 •• Engine Turbochargers and Intercoolers. The power output of an automobile engine is directly proportional to the mass of air that can be forced into the volume of the engine's cylinders to react chemically with gasoline. Many cars have a *turbocharger*, which compresses the air before it enters the engine, giving a greater mass of air per volume. This rapid, essentially adiabatic compression also heats the air. To compress it further, the air then passes through an *intercooler* in which the air exchanges heat with its surroundings at essentially constant pressure. The air is then drawn into the cylinders. In a typical installation, air is taken into the turbocharger at atmospheric pressure (1.01×10^5 Pa), density $\rho = 1.23\text{ kg/m}^3$, and temperature 15.0°C . It is compressed adiabatically to 1.45×10^5 Pa. In the intercooler, the air is cooled to the original temperature of 15.0°C at a constant pressure of 1.45×10^5 Pa. (a) Draw a

pV -diagram for this sequence of processes. (b) If the volume of one of the engine's cylinders is 575 cm^3 , what mass of air exiting from the intercooler will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C , what percentage increase in power is obtained by using the turbocharger and intercooler? (c) If the intercooler is not used, what mass of air exiting from the turbocharger will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C , what percentage increase in power is obtained by using the turbocharger alone?

MCAT-STYLE PASSAGE PROBLEMS

BIO Anesthetic Gases. One type of gas mixture used in anesthesiology is a 50%/50% mixture (by volume) of nitrous oxide (N_2O) and oxygen (O_2), which can be premixed and kept in a cylinder for later use. Because these two gases don't react chemically at or below 2000 psi, at typical room temperatures they form a homogeneous single gas phase, which can be considered an ideal gas. If the temperature drops below -6°C , however, N_2O may begin to condense out of the gas phase. Then any gas removed from the cylinder will initially be nearly pure O_2 ; as the cylinder empties, the proportion of O_2 will decrease until the gas coming from the cylinder is nearly pure N_2O .

19.63 In a test of the effects of low temperatures on the gas mixture, a cylinder filled at 20.0°C to 2000 psi (gauge pressure) is cooled slowly and the pressure is monitored. What is the expected pressure at -5.00°C if the gas remains a homogeneous mixture? (a) 500 psi; (b) 1500 psi; (c) 1830 psi; (d) 1920 psi.

19.64 In another test, the valve of a 500 L cylinder full of the gas mixture at 2000 psi (gauge pressure) is opened wide so that the gas rushes out of the cylinder very rapidly. Why might some N_2O condense during this process? (a) This is an isochoric process in which the pressure decreases, so the temperature also decreases. (b) Because of the rapid expansion, heat is removed from the system, so the internal energy and temperature of the gas decrease. (c) This is an isobaric process, so as the volume increases, the temperature decreases proportionally. (d) With the rapid expansion, the expanding gas does work with no heat input, so the internal energy and temperature of the gas decrease.

19.65 You have a cylinder that contains 500 L of the gas mixture pressurized to 2000 psi (gauge pressure). A regulator sets the gas flow to deliver 8.2 L/min at atmospheric pressure. Assume that this flow is slow enough that the expansion is isothermal and the gases remain mixed. How much time will it take to empty the cylinder? (a) 1 h; (b) 33 h; (c) 57 h; (d) 140 h.

19.66 In a hospital, pure oxygen may be delivered at 50 psi (gauge pressure) and then mixed with N_2O . What volume of oxygen at 20°C and 50 psi (gauge pressure) should be mixed with 1.7 kg of N_2O to get a 50%/50% mixture by volume at 20°C ? (a) 0.21 m^3 ; (b) 0.27 m^3 ; (c) 1.9 m^3 ; (d) 100 m^3 .

ANSWERS

Chapter Opening Question ?

(ii) The work done by a gas as its volume changes from V_1 to V_2 is equal to the integral $\int p \, dV$ between those two volume limits. If the volume of the gas contracts, the final volume V_2 is less than the initial volume V_1 and the gas does negative work. Propelling the locomotive requires that the gas do positive work, so the gas doesn't contribute to propulsion while contracting.

Key Example VARIATION Problems

VP19.5.1 (a) $5.75 \times 10^3 \text{ J}$ (b) $-3.20 \times 10^4 \text{ J}$ (c) $-1.85 \times 10^3 \text{ J}$ (d) 875 J

VP19.5.2 (a) 155 J (b) zero (c) -375 J (d) 875 J

VP19.5.3 (a) $W = -2.40 \times 10^2 \text{ J}$, $\Delta U = 1.80 \times 10^3 \text{ J}$

(b) $W = -7.20 \times 10^2 \text{ J}$, $Q = 1.08 \times 10^3 \text{ J}$

VP19.5.4 (a) $3.4 \times 10^6 \text{ J}$ (b) $3.4 \times 10^5 \text{ J}$ (c) $3.1 \times 10^6 \text{ J}$

VP19.6.1 (a) 998 J (b) 748 J (c) 599 J

VP19.6.2 (a) $5.24 \times 10^{-2} \text{ m}^3$ (b) (i) 327°C , $1.31 \times 10^4 \text{ J}$ (ii) 327°C , $1.31 \times 10^4 \text{ J}$ (iii) 927°C , $3.93 \times 10^4 \text{ J}$

VP19.6.3 (a) $4.50 \times 10^4 \text{ J}$ (b) $-2.25 \times 10^4 \text{ J}$ (c) $1.17 \times 10^5 \text{ J}$

VP19.6.4 (a) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{5}{2}nRT$, $W = 0$ (b) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{7}{2}nRT$, $W = nRT$ (c) $\Delta U = \frac{5}{2}nRT$, $Q = 0$, $W = -\frac{5}{2}nRT$

VP19.7.1 (a) $\frac{5}{3}$ (b) $6.41 \times 10^4 \text{ Pa}$ (c) 623 J

VP19.7.2 (a) $\frac{7}{5}$ (b) $5.79 \times 10^{-4} \text{ m}^3$ (c) $2.95 \times 10^6 \text{ Pa}$ (d) $-2.65 \times 10^3 \text{ J}$

VP19.7.3 (a) $3.17 \times 10^5 \text{ Pa}$ (b) $8.20 \times 10^4 \text{ Pa}$ (c) 178 K

(d) $7.94 \times 10^3 \text{ J}$

VP19.7.4 (a) $-3.38 \times 10^3 \text{ J}$ (b) 3.59 (c) 0.167

Bridging Problem

$$(a) W = nRT \ln \left[\frac{V_2 - nb}{V_1 - nb} \right] + an^2 \left[\frac{1}{V_2} - \frac{1}{V_1} \right]$$

(b) (i) $W = 2.80 \times 10^3 \text{ J}$, (ii) $W = 3.11 \times 10^3 \text{ J}$

(c) ideal gas, for which there is no attraction between molecules